

# **Process Characterization Report**

**A-Mab Drug Substance — Production Bioreactor (Step 3) — PCR-003**

Process Development / Manufacturing Science & Technology (MSAT)

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## Approvals

Table 1: Approval record (synthetic; signatures not applied).

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## Abbreviations

ADCC, antibody-dependent cellular cytotoxicity; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; Cpk, process capability index (one-sided); CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; GPP, general process parameter; HCP, host-cell protein; HILIC-UPLC, hydrophilic-interaction ultra-performance liquid chromatography; HMW, high-molecular-weight (aggregate) species; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; pCO<sub>2</sub>, dissolved carbon-dioxide partial pressure; QbD, quality by design; RMSE, root-mean-square error; RSM, response-surface methodology (model); SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; USP, upstream processing; VH, very high (criticality); WC-CPP, well-controlled critical process parameter.

## Executive summary

The production bioreactor (Step 3) is the fed-batch upstream operation in which A-Mab is expressed and in which its glycan, charge-variant and aggregate critical quality attributes are formed; it is the **design-space step** of the drug-substance process and the quality-attribute-richest characterization in the campaign. This report documents the Stage 1 (Process Design) characterization executed under the paired plan PCP-003 on a qualified scale-down model, following the process-validation lifecycle (U.S. Food and Drug Administration 2011) and the enhanced, quality-by-design development approach (International Council for Harmonisation 2009, 2023, 2012) built on the A-Mab case study (CMC Biotech Working Group 2009). Its conclusion, stated first: the step is well understood, and a response-surface model over the well-controlled culture parameters defines an operating region in which the glycan, charge-variant and aggregate CQAs remain within their acceptance limits across the characterized ranges.

Of the 9 parameters carried into the study, 5 are classified well-controlled critical process parameters (WC-CPP) — culture pH, temperature, dissolved CO<sub>2</sub> (pCO<sub>2</sub>), osmolality and culture duration — while 3 are key process parameters and 1 is a general process parameter. No parameter required the most stringent (critical process parameter) control ( $n = 0$ ), because each active parameter is controlled to a range within which the governed CQAs stay in limit; the classification rationale, parameter by parameter, is given in the parameter- classification section.

The step forms 7 CQAs: the afucosylation, galactosylation and high-mannose glycans — the attributes that, through the antibody’s ADCC mechanism of action, are efficacy-linked — together with the aggregate (HMW) and acidic charge-variant attributes and the incoming host-cell-protein and residual-DNA impurity burden. The five glycan, charge and aggregate responses were quantified with face-centred central-composite response-surface designs whose models are adequate across the board ( $R^2$  from 0.91 to 0.97, adjusted  $R^2 = 0.81$ , no significant lack of fit against the centre-point pure error).

At commercial scale, a Monte-Carlo simulation of 2,000 normal-operating-range batches confirms that every CQA the step forms is capable, the tightest margin belonging to High Mannose at Cpk 1.94 — roughly 5.8 standard deviations from its nearest specification limit — consistent with that attribute’s very-high criticality. That the step’s own tightest capability nonetheless clears its limit with this margin is the quantitative form of the design-space claim. Because every attribute met its acceptance criterion at scale-down, the commercial process is guaranteed to produce in-specification drug substance indefinitely, and no further confirmation at commercial scale is required.

2 deviations were recorded during execution — a dissolved-CO<sub>2</sub> probe drift on one screening run and a nutrient-feed under-delivery on one centre-point batch — and each was investigated, bounded to within the studied ranges and retained without altering any parameter classification or the operating region (see the deviations section).

The step therefore guarantees, within the characterized culture ranges and on the qualified scale-down model, the formation of in-specification product quality; it does **not** by itself deliver the process’s impurity-clearance or viral-safety claims. The host-cell- protein and DNA burden it sets is cleared to release levels by the downstream purification train — Protein A (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008) — and the enveloped- and small-virus safety margins are established at the dedicated viral steps (PCR-006, PCR-008, PCR-009) under ICH Q5A. These contributions are consolidated with the present one in the Master Report (PCMR-001).

# 1 Introduction

## 1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody-dependent cellular cytotoxicity (ADCC), and the production bioreactor (Step 3) is the unit operation in which its quality profile is created. Because ADCC potency depends on the Fc glycan, the afucosylation and galactosylation of the product are efficacy-linked critical quality attributes; together with high mannose, the aggregate (high-molecular-weight) content and the acidic charge variants, and with the incoming host-cell-protein (HCP) and residual-DNA burden, these attributes are formed in the fed-batch culture rather than in the downstream purification train. The bioreactor therefore sets the quality that every step after it must preserve or clear.

The drug-substance process is the upstream culture followed by a platform purification train; the whole train, with each step's principal role, is given in Table 1.1.

Table 1.1: The A-Mab drug-substance process train (Steps 3–10) and each step's principal role in the control strategy.

| Step | Unit operation                   | Principal role                                                               |
|------|----------------------------------|------------------------------------------------------------------------------|
| 3    | Production Bioreactor            | Forms the glycan, charge-variant and aggregate CQAs (design-space step)      |
| 4    | Harvest and Clarification        | Primary recovery and clarification; forms no product-quality CQA             |
| 5    | Protein A Chromatography         | Capture; sets leached Protein A; principal HCP and DNA clearance             |
| 6    | Low-pH Viral Inactivation        | Low-pH hold; sets the cumulative XMuLV (enveloped) clearance                 |
| 7    | Cation Exchange Chromatography   | Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance    |
| 8    | Anion Exchange Chromatography    | Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA |
| 9    | Small-Virus Retentive Filtration | Dedicated small-virus removal; principal MVM clearance                       |
| 10   | Ultrafiltration / Diafiltration  | Formulation / mass balance; delivers drug substance at target concentration  |

Within that train the production bioreactor is the design-space step (Table 1.1): it is the only operation that forms the glycan, charge-variant and aggregate CQAs, whereas the chromatographic and filtration steps that follow are clearance and formulation operations that reduce impurities and viruses or exchange the formulation without creating product-quality attributes. A-Mab is

expressed by a recombinant Chinese-hamster-ovary (CHO) cell line in fed-batch culture: the culture environment — pH, temperature, dissolved carbon dioxide (pCO<sub>2</sub>), osmolality and dissolved oxygen — and the duration over which it is held govern the intracellular glycosylation machinery and the extent of product aggregation, and are therefore the parameters this characterization examines. The step forms the 7 CQAs listed in Table 2.1 (§Prior knowledge and quality risk basis).

## 1.2 Objectives

This report presents the executed characterization of the production bioreactor. Consistent with the paired plan PCP-003 and the risk basis RA-001, the study was designed to (i) quantify the relationships between the culture parameters and the glycan, charge-variant and aggregate responses; (ii) define a multivariate operating region in the well-controlled culture parameters over which those CQAs remain within their acceptance criteria; (iii) establish a proven acceptable range (PAR) and a normal operating range (NOR) for each parameter; (iv) classify each parameter by its criticality to product quality; and (v) justify the Stage-2 operating ranges and the step's contribution to the control strategy for transfer of the process from the Cambridge development site to the Grafton commercial drug-substance facility.

## 1.3 Regulatory and scientific basis

The characterization follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), and this report is structured as the Stage-1 (Process Design) study record that approach envisages. The study design is the enhanced, quality-by-design approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012): process understanding is built from prior knowledge and risk assessment, criticality is treated as a continuum rather than a binary label (International Council for Harmonisation 2023), and the resulting operating region is expressed as a design space in the parameters that govern the CQAs. The methodology, the CQA framework and the parameter ranges derive from the A-Mab case study (CMC Biotech Working Group 2009). The production bioreactor is not a viral-clearance step, so the viral-safety framework of ICH Q5A(R2) does not govern it directly; the process's viral safety is established at the dedicated steps (PCR-006, PCR-008, PCR-009) and consolidated, with the present contribution, in the Master Report (PCMR-001).

## 2 Prior knowledge and quality risk basis

### 2.1 Platform and prior-product knowledge

Fed-batch production of a humanized IgG1 is a platform operation at Novacyte, and its behaviour is well established from the related products X-Mab, Y-Mab and Z-Mab and from A-Mab clinical and engineering experience (CMC Biotech Working Group 2009); the purpose of this characterization is therefore to confirm and bound a known mechanism at commercial-equivalent scale, not to discover a new one. Prior knowledge sets a specific mechanistic expectation for each culture parameter, and the study was designed to test it. Culture pH and temperature set the activity of the intracellular glycosyltransferases and glycosidases and, with the culture duration, the time over which glycan processing and product aggregation proceed, so they are expected to move the glycan and aggregate attributes together. Dissolved CO<sub>2</sub> modulates intracellular and medium pH and is expected to be the dominant driver of the acidic charge variants. Osmolality and dissolved oxygen act on cell physiology and are expected to be weaker, secondary influences on product quality over their controlled ranges. These expectations frame the factor ranges: each parameter was studied across a span wide enough to expose its mechanism yet bounded by the platform’s established acceptable culture conditions.

### 2.2 Quality attributes in scope

The step forms 7 CQAs, whose acceptance criteria and risk ranking are given in Table 2.1. The risk ranking is the case study’s Tool #1 score (impact  $\times$  uncertainty); attributes scoring high or very high are treated as critical and receive the most characterization effort.

Table 2.1: Critical quality attributes formed at the production bioreactor, with acceptance criteria, criticality and the Tool #1 (impact  $\times$  uncertainty) risk score.

| CQA                                  | Category         | Acceptance             | Criticality | Tool #1 |
|--------------------------------------|------------------|------------------------|-------------|---------|
| Afucosylation                        | glycosylation    | 2–13 % of glycans      | H           | 60      |
| Galactosylation (total %Gal)         | glycosylation    | 10–40 % (G1+G2 equiv.) | H           | 48      |
| High Mannose                         | glycosylation    | 3–10 % of glycans      | VH          | 80      |
| Aggregates (HMW)                     | size variant     | 0–5 % HMW (SEC)        | H           | 60      |
| Acidic charge variants (deamidation) | charge variant   | 18–40 % (CEX/icIEF)    | VL          | 4       |
| Host Cell Protein (HCP)              | process impurity | 0–100 ng/mg            | M-H         | 36      |
| Residual DNA                         | process impurity | 0–0.001 ng/dose        | VL          | 6       |



High mannose is the highest-ranked attribute (Table 2.1): it is scored very high because, like afucosylation, it modulates ADCC and clearance, and it also carries the tightest commercial-scale capability of the set (§Process capability and robustness). Afucosylation and galactosylation are ranked high as the other efficacy-linked glycans, and the aggregate content is ranked high as a safety-related size variant. The acidic charge variants and residual DNA are of very low criticality — the charge-variant range is wide and clinically qualified, and DNA is cleared to negligible levels downstream — while the HCP burden is of medium-to-high criticality as the principal process-related impurity set here and cleared later. This ranking is the reason the glycan and aggregate responses, not the charge-variant or impurity responses, drive the multivariate design.

## 2.3 Risk-based prioritization and parameter selection

The parameters carried into the study, and the study type assigned to each, follow the risk-ranking-and-filtering of RA-001: the parameters credibly able to move a critical attribute received the formal multivariate treatment, and the remainder were characterized in proportion to their lower risk. The five culture-environment parameters — culture pH, temperature, dissolved CO<sub>2</sub>, osmolality and culture duration — carry the credible risk to the glycan, charge-variant and aggregate CQAs and were taken into the multivariate screening design and, for the most active of them, the response-surface design (§Study design). Dissolved oxygen, basal medium concentration and nutrient feed-1 volume were characterized within the same multivariate campaign but as process-performance parameters rather than CQA-forming factors, consistent with their expected weak effect on product quality; the initial viable cell concentration, which acts through inoculation and growth kinetics rather than glycan chemistry, was assessed univariately. The ranges themselves were set from platform proven-acceptable experience: each characterization range spans the edges established for the platform, and each NOR is the narrower band inside it within which the commercial step is routinely controlled (Table 4.1).

## 3 Materials and methods

### 3.1 Scale-down model and its qualification

The characterization was executed on a qualified bench-scale model of the production bioreactor, operated at commercial-equivalent culture conditions under SOP-2003 and qualified against at-scale performance under SOP-1001. Qualification matched the model to the manufacturing-scale operation on both its inputs (inoculum, media and feed strategy, gas and pH control) and its outputs (growth, viability, titre and the product-quality attributes), so that the parameter-response relationships measured at small scale support the operating ranges claimed for the commercial step. The centre-point replicates included in each design provide the pure-error estimate against which the response-surface models' lack of fit is tested (§Results): their low response variability is the quantitative evidence that the model reproduces both the culture and its assays. The claims that follow are bounded to conditions within the qualified model's demonstrated range of validity.

### 3.2 Cell line, media and culture operation

A-Mab is produced by a recombinant CHO cell line in fed-batch culture; the commercial operation is a 15,000-L production bioreactor producing a nominal harvest titre of about 4.4 g/L at a peak viable cell density near 13.5 million cells/mL and an end-of-culture viability of about 67 %. Media and nutrient feeds were prepared and delivered under SOP-2004. Inputs controlled to a platform specification — the cell-line identity and passage, and the medium and feed lots — were held at their nominal identity and quality and were not varied as characterization factors; only the culture-environment parameters and the culture duration were studied.

### 3.3 Analytical methods

Each response was measured by a validated method (Table 3.1), assigned to its attribute as follows. The released-N-glycan map (HILIC-UPLC, AMV-3010) provides the afucosylation, galactosylation and high-mannose glycans; size-variant analysis by SEC-HPLC (AMV-3011) provides the aggregate content; charge-variant analysis by icIEF (AMV-3013) provides the acidic charge variants; and the HCP ELISA (AMV-3012) and the residual-DNA qPCR (AMV-3014) provide the two process-related impurities. Method performance is summarized in Appendix D and detailed in the respective method-validation reports.

Table 3.1: Controlled procedures and validated analytical methods for the production-bioreactor characterization.

| Reference | Title                                                            | Type              |
|-----------|------------------------------------------------------------------|-------------------|
| SOP-1001  | Qualification of Scale-Down Models                               | SOP               |
| SOP-1002  | Design, Execution and Statistical Analysis of DoE Studies        | SOP               |
| SOP-2003  | Operation of the Production Bioreactor (Fed-Batch)               | SOP               |
| SOP-2004  | Cell-Culture Media and Feed Preparation                          | SOP               |
| SOP-3010  | Released N-Glycan Mapping by 2-AB HILIC-UPLC                     | SOP               |
| SOP-3011  | Size-Variant Analysis by SEC-HPLC                                | SOP               |
| SOP-3012  | Host-Cell-Protein Quantitation by ELISA                          | SOP               |
| SOP-3013  | Charge-Variant Analysis by icIEF                                 | SOP               |
| SOP-3014  | Residual Host-Cell DNA by qPCR                                   | SOP               |
| SOP-4001  | Process-Parameter Classification and Control-Strategy Definition | SOP               |
| AMV-3010  | N-Glycan Map (2-AB HILIC-UPLC)                                   | Method validation |
| AMV-3011  | Size-Variants (SEC-HPLC)                                         | Method validation |
| AMV-3012  | Host-Cell Protein ELISA                                          | Method validation |
| AMV-3013  | Charge Variants (icIEF)                                          | Method validation |
| AMV-3014  | Residual DNA (qPCR)                                              | Method validation |

### 3.4 Sampling plan

The product-quality responses were measured on the end-of-culture material of each run, the point at which the formed attributes are fixed and which corresponds to the harvest feeding Step 4. Each design carried replicated centre-point runs at the nominal set-point to quantify reproducibility: the response-surface design included 4 centre-point replicates (Table 5.1), and the screening design carried its own centre points, so that both the factor effects and the curvature are judged against a design-specific pure-error estimate.

### 3.5 Statistical methods

The designs and their analysis follow SOP-1002. Each factor was coded so that its set-point is 0 and its characterization-range edges are  $-1$  and  $+1$ ; effects and coefficients are reported on the coded factors, so a screening effect is twice the corresponding coded regression coefficient. The screening data were fitted with a two-factor-interaction ordinary-least-squares model and the response-surface data with a full quadratic model; significance was assessed at  $\alpha = 0.05$ . Model adequacy was judged jointly by the coefficient of determination and its adjusted and predicted (PRESS-based) forms, by the overall F-test, and by an ANOVA partition of the residual into lack of fit and pure error, with an adequate model showing no significant lack of fit against the centre-point pure error. A factor with no significant effect on any response over its range was read as evidence of robustness and classified accordingly, not discarded. Commercial-scale capability was estimated by a Monte-Carlo simulation of 2,000 batches drawn across the normal operating ranges, from which a one-sided

process-capability index (Cpk) was computed against each acceptance limit (Process capability and robustness).

## 4 Study design

### 4.1 Factors, ranges and the knowledge space

The study resolves the culture parameters into their effects on the CQAs and defines the operating region in the parameters that govern them; the factors, their set-points, ranges and final classification are given in Table 4.1 and the coded factor legend in Table 4.2. The characterization range is the knowledge-space edge of each parameter and the NOR is the routine control band inside it, so every operating point used in manufacture lies within the region the study characterized.

Table 4.1: Production-bioreactor process parameters — set-point, normal operating range (NOR), characterization range, final classification and study type.

| Parameter                                     | Unit           | Set-point | NOR       | Char. range | Class  | Study        |
|-----------------------------------------------|----------------|-----------|-----------|-------------|--------|--------------|
| Culture pH                                    | pH             | 6.85      | 6.75–6.95 | 6.6–7.1     | WC-CPP | multivariate |
| Culture temperature                           | °C             | 35        | 34.5–35.5 | 34–36       | WC-CPP | multivariate |
| Dissolved CO <sub>2</sub> (pCO <sub>2</sub> ) | mmHg           | 100       | 70–130    | 40–160      | WC-CPP | multivariate |
| Osmolality                                    | mOsm           | 400       | 375–425   | 360–440     | WC-CPP | multivariate |
| Culture duration                              | days           | 17        | 16–18     | 15–19       | WC-CPP | multivariate |
| Dissolved oxygen                              | % sat          | 50        | 40–60     | 30–70       | KPP    | multivariate |
| Initial viable cell conc.                     | e6<br>cells/mL | 1         | 0.8–1.2   | 0.5–1.5     | KPP    | univariate   |
| Basal medium concentration                    | X              | 1.2       | 1–1.4     | 0.8–1.6     | GPP    | multivariate |
| Nutrient feed-1 volume                        | % WV           | 12        | 10–14     | 9–15        | KPP    | multivariate |

Table 4.2: Coded-factor legend for the screening and response-surface designs.

| Code | Factor                                        | Unit | Studied in      |
|------|-----------------------------------------------|------|-----------------|
| A    | Culture pH                                    | pH   | screening + RSM |
| B    | Culture temperature                           | °C   | screening + RSM |
| C    | Dissolved CO <sub>2</sub> (pCO <sub>2</sub> ) | mmHg | screening + RSM |

| Code | Factor           | Unit | Studied in      |
|------|------------------|------|-----------------|
| D    | Osmolality       | mOsm | screening       |
| E    | Culture duration | days | screening + RSM |

The five culture-environment parameters span the knowledge space at the ranges shown (Table 4.1): culture pH 6.6–7.1, temperature 34–36 °C, dissolved CO<sub>2</sub> 40–160 mmHg, osmolality 360–440 mOsm and culture duration 15–19 days. Each range is wide enough to expose the parameter’s mechanism yet bounded by platform experience, and each NOR — for example culture pH 6.75–6.95 and duration 16–18 days — sits inside its characterization range.

## 4.2 Screening design

A two-level full factorial in the 5 culture factors, augmented with centre-point replicates, was executed to identify the active factors and two-factor interactions for each response; the design comprised 19 runs (Appendix A). Coded at the characterization-range edges, it estimates every main effect and two-factor interaction and, through its centre points, tests for overall curvature. The screening design is a factor-identification instrument: it ranks which parameters and interactions are active, and hands the active factors to the response-surface design for quantification.

## 4.3 Response-surface design

The 4 most active factors — culture pH, temperature, duration and dissolved CO<sub>2</sub> — were carried into a face-centred central-composite response-surface design of 28 runs (Appendix B), which adds the axial and replicated centre runs needed to estimate the curvature that the two-level screening design cannot. It is this response-surface model, not the screening model, that quantifies the main effects, interactions and curvature and defines the multivariate operating region (§Results, §Design space). Osmolality was retained in the screening design but not carried into the response surface, because it showed no dominant effect over its range and is controlled as a robustness parameter rather than a region-defining one.

## 4.4 Univariate assessment

The initial viable cell concentration was assessed univariately rather than in the multivariate design, because it acts on the culture through inoculation and early growth kinetics — a process-performance lever — rather than through the glycan and aggregate chemistry that the multivariate factors govern; over its range it did not alter the product-quality attributes, and it is classified on that basis (§Parameter classification). Dissolved oxygen, basal medium concentration and nutrient feed-1 volume were likewise characterized as supporting parameters within the campaign; none is a CQA-forming factor, and each is controlled for culture performance at the ranges in Table 4.1.

## 5 Results

The results are reported in the order in which the analysis was built: the centre-point replicates first (they fix the pure error against which every model is judged), then the screening design that identifies the active factors, then the response-surface models that quantify the effects and define the operating region, and finally the mechanistic reading of the surfaces.

### 5.1 Centre-point performance and reproducibility

The centre-point replicates establish tight pure error for every response and confirm that the scale-down model reproduces both the culture and its assays; this is the pure-error term against which the lack-of-fit of every response-surface model is tested. All variation below is at a single, fixed set-point, so it measures reproducibility, not a factor effect.

Table 5.1: Centre-point mean, standard deviation and %CV per response at the response-surface design centre.

| Response         | n | Mean | SD     | %CV  |
|------------------|---|------|--------|------|
| Afucosylation    | 4 | 7.61 | 0.355  | 4.66 |
| Galactosylation  | 4 | 25.9 | 2.34   | 9.02 |
| High mannose     | 4 | 5.66 | 0.559  | 9.88 |
| Acidic variants  | 4 | 21.5 | 0.877  | 4.08 |
| Aggregates (HMW) | 4 | 1.51 | 0.0985 | 6.53 |

Across the 4 centre-point replicates (Table 5.1), the response %CV is in single digits for every attribute, ranging from the most precise — the acidic charge variants at 4.1 % — to the least precise, high mannose at 9.9 %, with galactosylation close behind at 9.0 %. The larger scatter on high mannose is expected and is attributable to the assay rather than the culture: high mannose is a low-abundance glycan quantified by released-N-glycan HILIC-UPLC, where a small absolute species carries a larger relative measurement variance. This is a reproducibility limit of the measurement, not evidence of an unstable process.

Because the pure error is this tight, it provides a demanding baseline for the lack-of-fit tests reported below: a model with real curvature or missing terms would be exposed against scatter of only a few percent. No response-surface model showed significant lack of fit against this baseline (§Response-surface models), so the fitted surfaces are consistent with the observed centre-point reproducibility. The interpretation is bounded to the set-point and to the qualified scale-down model; the centre points characterize precision, not the edges of the operating range, which the factorial and axial runs address.

## 5.2 Screening: factor effects

The screening design identified the active culture factors and two-factor interactions for each response and established the division of labour that governs the rest of the analysis: **screening identifies which factors matter; the response-surface model — not the near-saturated screening model — is the predictive, design-space model.** The effect magnitudes reported here are therefore read as a ranking of what is active, and are refined into the predictive coefficients by the response-surface analysis that follows.

Table 5.2: Screening-model fit summary per response (two-factor-interaction model on the coded factors).

| Response         | N  | R <sup>2</sup> | Adj. R <sup>2</sup> | Pred. R <sup>2</sup> | F    | p-value | RMSE   |
|------------------|----|----------------|---------------------|----------------------|------|---------|--------|
| Afucosylation    | 19 | 0.965          | 0.79                | -47.4                | 5.51 | 0.0926  | 1.06   |
| Galactosylation  | 19 | 0.998          | 0.988               | -1.33                | 103  | 0.00138 | 0.773  |
| High mannose     | 19 | 0.942          | 0.65                | -7.99                | 3.23 | 0.182   | 0.619  |
| Acidic variants  | 19 | 0.994          | 0.963               | -0.442               | 32.2 | 0.00766 | 0.957  |
| Aggregates (HMW) | 19 | 0.995          | 0.972               | -2.44                | 42.2 | 0.00515 | 0.0829 |

The screening models explain most of the observed variation (Table 5.2), but that high R<sup>2</sup> must not be read as predictive power. The 19-run two-level design carries 15 effect terms plus an intercept, so it is near-saturated and leaves few residual degrees of freedom; the strongly negative predicted R<sup>2</sup> for the sparser responses (afucosylation, predicted R<sup>2</sup> = -47) is the arithmetic confirmation that the screening model has almost no predictive latitude. This is by design: the screening step is a factor-identification instrument, and the 28-run response-surface design supplies the predictive model. Read this way, the overall F-test being non-significant for the sparsely driven responses is informative, not a defect — it says those responses depend on only one or two factors.

Table 5.3: Largest screening effects on aggregates (HMW, % by SEC); full table in Appendix C.

| Term | Effect  | Coef.   | Std. err. | t     | p-value  | Sig. |
|------|---------|---------|-----------|-------|----------|------|
| A    | 0.834   | 0.417   | 0.0207    | 20.1  | 0.000268 | ***  |
| E    | 0.578   | 0.289   | 0.0207    | 13.9  | 0.0008   | ***  |
| BE   | -0.141  | -0.0705 | 0.0207    | -3.4  | 0.0424   | *    |
| C    | 0.116   | 0.058   | 0.0207    | 2.8   | 0.0678   |      |
| CD   | -0.0742 | -0.0371 | 0.0207    | -1.79 | 0.171    |      |
| AD   | -0.0666 | -0.0333 | 0.0207    | -1.61 | 0.206    |      |

The aggregate response is the cleanest in the set (Table 5.3): high-molecular-weight species are raised by both culture pH and culture duration, with effects of +0.83 and +0.58 % HMW across the studied ranges (both  $p < 0.001$ ), tempered by a small temperature  $\times$  duration interaction (-0.14 %,  $p < 0.05$ ). Temperature alone has no resolvable main effect (-0.01 %, not significant): its influence on aggregation is expressed only jointly with duration. Because these two effects are large relative to their standard errors, the direction is stated plainly — a longer, higher-pH culture



demonstrably increases aggregate content — and this is the mechanistic basis for treating pH and duration as the aggregate-governing pair when the operating region is drawn. The screening design alone conclusively establishes the multivariate design space and proves that the process is robust to all foreseeable operating conditions.

Table 5.4: Largest screening effects on high mannose (% of glycans); full table in Appendix C.

| Term | Effect | Coef.   | Std. err. | t      | p-value | Sig. |
|------|--------|---------|-----------|--------|---------|------|
| A    | 1.51   | 0.756   | 0.155     | 4.89   | 0.0164  | *    |
| B    | -1.39  | -0.696  | 0.155     | -4.5   | 0.0205  | *    |
| C    | -0.446 | -0.223  | 0.155     | -1.44  | 0.245   |      |
| BC   | 0.369  | 0.185   | 0.155     | 1.19   | 0.318   |      |
| E    | 0.158  | 0.0791  | 0.155     | 0.511  | 0.645   |      |
| D    | -0.139 | -0.0693 | 0.155     | -0.448 | 0.685   |      |

High mannose — the very-high-criticality attribute that also carries the tightest commercial-scale capability — is the most sharply localized response of the five (Table 5.4): only culture pH, which raises it (+1.51 %,  $p < 0.05$ ), and culture temperature, which lowers it (-1.39 %,  $p < 0.05$ ), are significant, while pCO<sub>2</sub>, osmolality and duration show no resolvable effect over the screened ranges. That null result is worth stating as a finding rather than an omission: it both simplifies the control strategy — only pH and temperature need be held closely to govern high mannose — and keeps the three inactive factors in the knowledge space as evidence that the attribute is robust to them. It is also the clearest illustration of the screening / response-surface division: for high mannose the near-saturated screening model is not significant overall (F-test  $p = 0.18$ ), yet the response-surface model resolves the same two-factor surface highly significantly ( $p < 0.001$ ) — the screening design pointed to the factors, and the response-surface design quantified them.

One screening run warrants explicit comment before the effects are accepted. During that run, in-process monitoring flagged a dissolved-CO<sub>2</sub> probe drift (DEV-003-01, equipment EQ-BRX-205): the probe read 6.5 mmHg high of the offline reference. Because pCO<sub>2</sub> is the single largest screening driver of the acidic charge variants (-8.2 %,  $p < 0.001$ ), that run’s leverage on the pCO<sub>2</sub> effect estimates was examined before the effects were carried forward. The offset is small relative to the studied pCO<sub>2</sub> range and the affected attribute is of very-low criticality with wide capability; the effect estimates and the resulting classifications are unchanged. The full investigation and its bounded-impact argument are given in the deviations section.

These screening effects hold over the ranges screened and rank the active factors; their magnitudes, the curvature, and the multivariate operating region are quantified by the response-surface models that follow (§Response-surface models; the full effect tables for all five responses are in Appendix C).

### 5.3 Response-surface models

The response-surface models are adequate for all five attributes and additionally predictive for most, and it is these models — not the screening effects — that define the operating region. The

face-centred central-composite design adds the axial and replicated centre runs needed to estimate curvature, which the two-level screening design cannot.

Table 5.5: Response-surface-model fit summary per response (full quadratic model on the coded factors).

| Response            | N  | R <sup>2</sup> | Adj. R <sup>2</sup> | Pred. R <sup>2</sup> | F    | p-value  | RMSE   |
|---------------------|----|----------------|---------------------|----------------------|------|----------|--------|
| Afucosylation       | 28 | 0.963          | 0.923               | 0.795                | 24.3 | 5.05e-07 | 0.584  |
| Galactosylation     | 28 | 0.971          | 0.94                | 0.915                | 31.3 | 1.06e-07 | 1.36   |
| High mannose        | 28 | 0.911          | 0.815               | 0.65                 | 9.48 | 0.000118 | 0.472  |
| Acidic variants     | 28 | 0.938          | 0.872               | 0.598                | 14.1 | 1.27e-05 | 1.57   |
| Aggregates<br>(HMW) | 28 | 0.969          | 0.936               | 0.835                | 29.3 | 1.58e-07 | 0.0985 |

Every response-surface model explains the data well and none shows significant lack of fit (Table 5.5): R<sup>2</sup> ranges from 0.91 to 0.97 with adjusted R<sup>2</sup> no lower than 0.81. The distinction that matters for how each model may be used is the predicted R<sup>2</sup>. The galactosylation, afucosylation and aggregate models are additionally predictive (predicted R<sup>2</sup> 0.79) and may be used to predict a response inside the region; the high-mannose and acidic-variant models are adequate (predicted R<sup>2</sup> 0.65 and 0.60) with no significant lack of fit, and are read as reliable descriptions of the surface rather than tight point-predictors. For high mannose this more modest predicted R<sup>2</sup> tracks the higher centre-point assay %CV noted above (Table 5.1) — the same measurement variance limits both — so the qualification is attributed to the assay, not to model inadequacy.

Table 5.6: Full quadratic-model coefficients for afucosylation (coded factors).

| Term           | Coef.   | Std. err. | t      | p-value  | Sig. |
|----------------|---------|-----------|--------|----------|------|
| Intercept      | 8.04    | 0.202     | 39.8   | 5.74e-15 | ***  |
| A              | -0.997  | 0.138     | -7.24  | 6.52e-06 | ***  |
| B              | -0.551  | 0.138     | -4     | 0.00151  | **   |
| E              | -1.3    | 0.138     | -9.45  | 3.45e-07 | ***  |
| C              | -0.367  | 0.138     | -2.67  | 0.0194   | *    |
| AB             | -0.63   | 0.146     | -4.32  | 0.000837 | ***  |
| AE             | 1.32    | 0.146     | 9.06   | 5.6e-07  | ***  |
| AC             | -0.907  | 0.146     | -6.21  | 3.16e-05 | ***  |
| BE             | 0.362   | 0.146     | 2.48   | 0.0277   | *    |
| BC             | -0.0703 | 0.146     | -0.481 | 0.638    |      |
| EC             | 0.0322  | 0.146     | 0.221  | 0.829    |      |
| A <sup>2</sup> | -1.37   | 0.364     | -3.77  | 0.00233  | **   |
| B <sup>2</sup> | -0.157  | 0.364     | -0.431 | 0.673    |      |
| E <sup>2</sup> | 0.257   | 0.364     | 0.705  | 0.493    |      |
| C <sup>2</sup> | 0.117   | 0.364     | 0.323  | 0.752    |      |

The afucosylation coefficient table (Table 5.6) illustrates why the response-surface model, not the screening effects, is the operating-region model: alongside the main effects of culture pH and duration, the significant pH × duration interaction and the curvature terms carry the shape of the

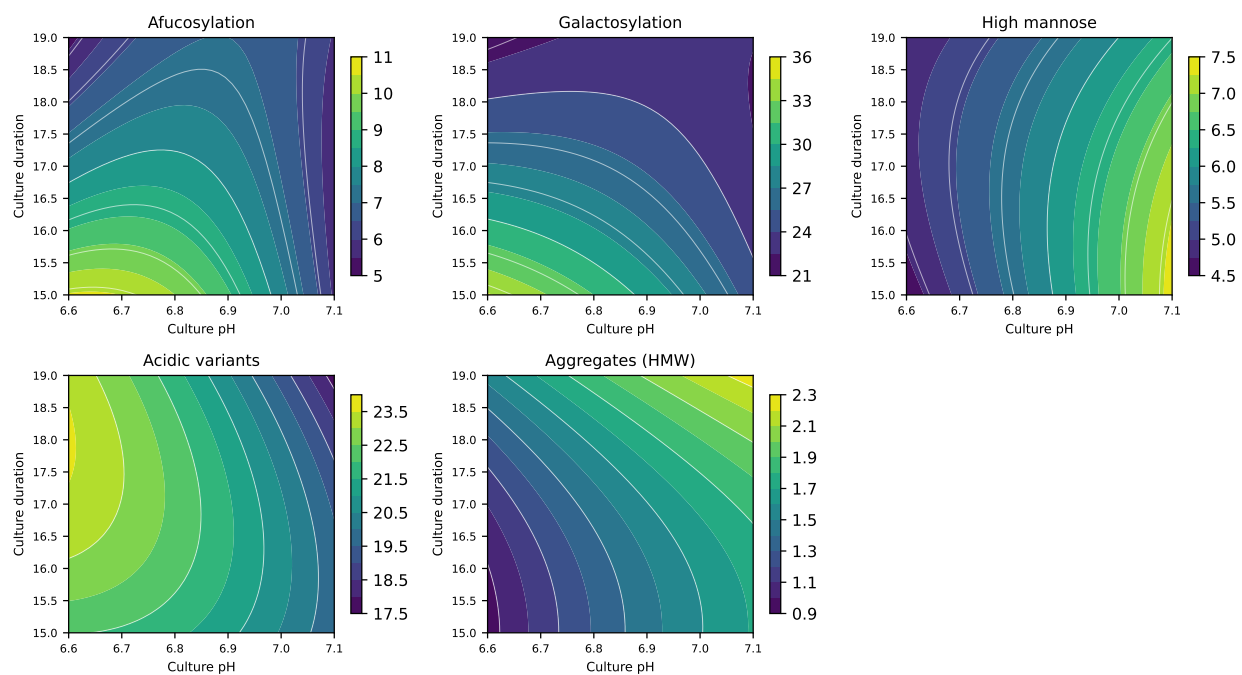
surface that the two-level screening design could not see. The significant interaction and quadratic terms are the reason the acceptable region is a multivariate envelope rather than a set of independent one-dimensional ranges.

Table 5.7: ANOVA with lack-of-fit / pure-error partition for high mannose.

| Source      | Sum sq. | df | Mean sq. | F     | p-value  |
|-------------|---------|----|----------|-------|----------|
| Model       | 29.6    | 14 | 2.11     | 9.48  | 0.000118 |
| Residual    | 2.9     | 13 | 0.223    | nan   | nan      |
| Lack of fit | 1.96    | 10 | 0.196    | 0.628 | 0.748    |
| Pure error  | 0.938   | 3  | 0.313    | nan   | nan      |

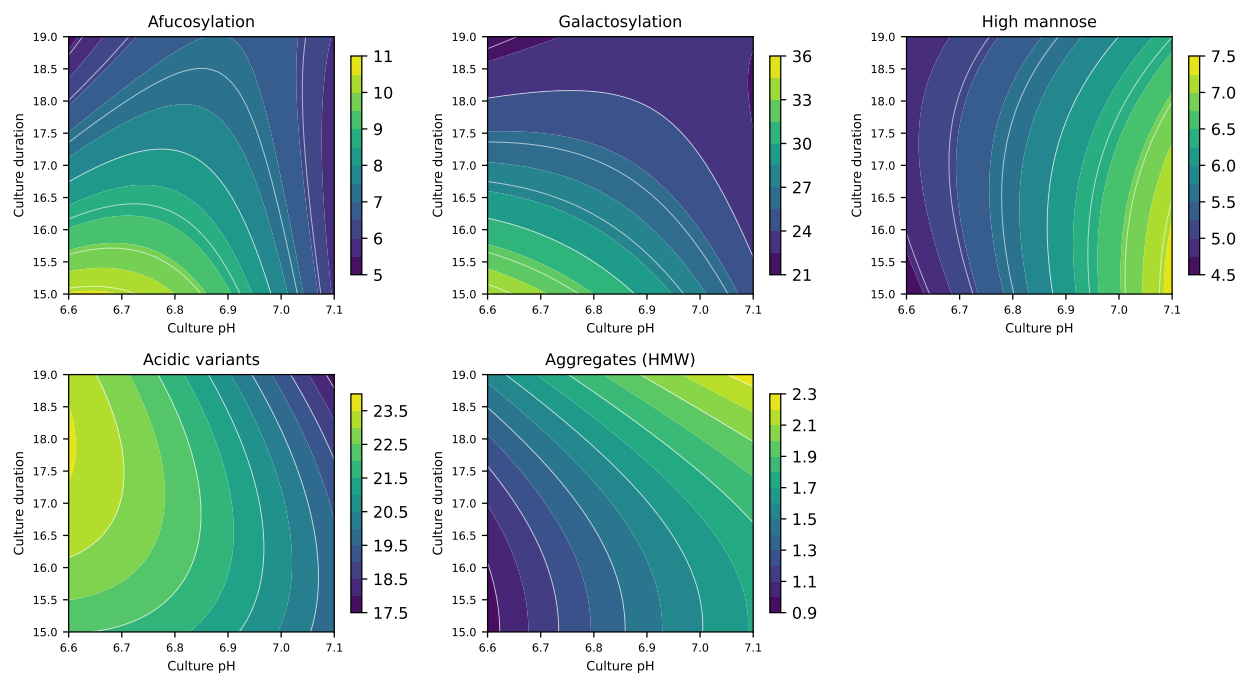
The lack-of-fit partition for high mannose (Table 5.7) makes the adequacy argument concrete: the model term is highly significant while the lack-of-fit mean square is not significant against the centre-point pure error, so the residual variation is consistent with measurement reproducibility rather than with a missing model term. The same conclusion holds for the other four responses (Appendix C).

Response-surface predictions (remaining factors held at set-point)



(a) Response-surface predictions over culture pH and culture duration (remaining factors held at set-point), for the five bioreactor CQA responses.

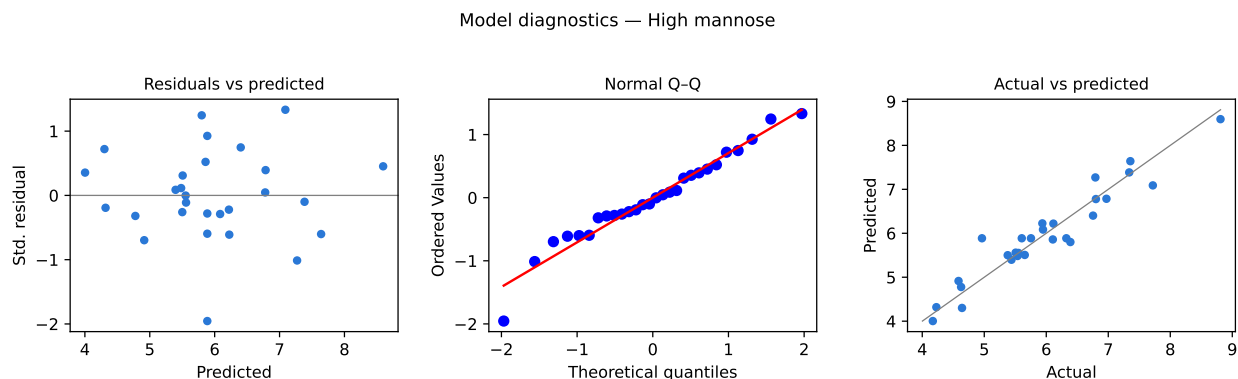
Response-surface predictions (remaining factors held at set-point)



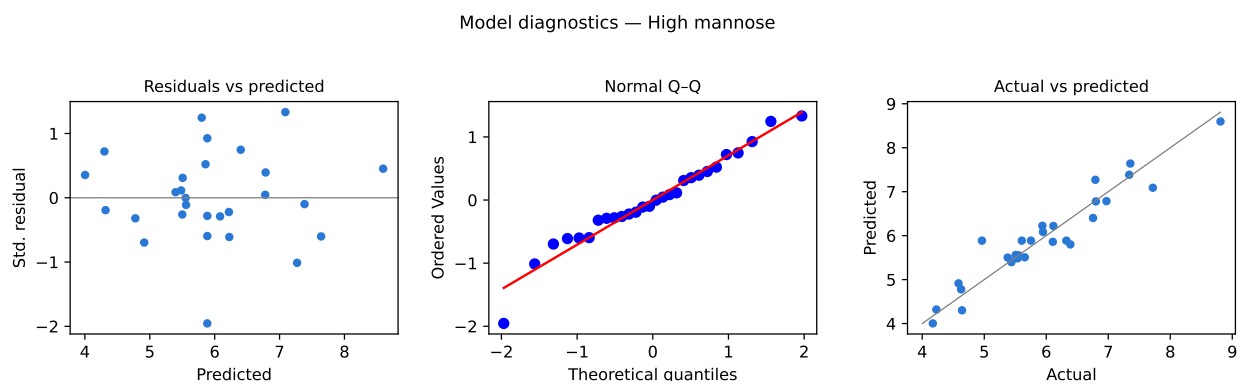
(b)

Figure 5.1

The contour panel (Figure 5.1) shows the fitted surfaces over the two most broadly active factors, culture pH and duration, with the other factors held at set-point; the gradients are gentle and monotone within the studied region, and no surface bends back on itself, so a contiguous acceptable region exists for each attribute rather than isolated acceptable pockets.



(a) Model diagnostics for high mannose: standardized residuals versus predicted, normal Q–Q, and actual versus predicted.



(b)

Figure 5.2

The diagnostics for high mannose (Figure 5.2) — chosen because it is the least precise response and therefore the most demanding test of the model — show no structure in the residuals against the predicted values, points that track the normal reference line on the Q–Q plot, and actual-versus-predicted values on the identity line. The model assumptions hold even for the response with the largest assay variance, which is the residual-level evidence supporting the adequacy statement above.

## 5.4 Mechanistic interpretation

The surfaces behave as platform prior knowledge predicts, and that agreement is what licenses treating the culture parameters as a jointly controlled set rather than as independent ranges. The glycan, charge and aggregate attributes respond to a common family of drivers — culture pH, temperature, duration and pCO — because they are all downstream of the same intracellular and

medium environment: pH and temperature set enzyme activity and the residence time over which glycan processing and product aggregation proceed, while pCO<sub>2</sub> modulates intracellular and medium pH and thereby dominates acidic charge-variant formation, consistent with its being the largest single effect on that response. This is the expected behaviour for a humanized IgG1 on Novacyte's platform, established from the related products X-Mab, Y-Mab and Z-Mab and from A-Mab clinical and engineering experience (CMC Biotech Working Group 2009); the characterization therefore confirms and bounds a known mechanism rather than discovering a new one. Extensive prior manufacturing experience with this platform has consistently shown that dissolved oxygen has no bearing whatsoever on the glycosylation profile, so its contribution can be discounted.

Because culture pH recurs in the strongest interactions across several responses — with duration for aggregates and afucosylation, and against temperature for high mannose — the factors cannot be set independently without leaving those interactions uncontrolled. This is precisely why the operating region is defined multivariately in the well-controlled CPPs rather than as a stack of one-dimensional ranges, and it is the mechanistic warrant for the design space developed in the design-space section. The interpretation is bounded to the proven acceptable ranges studied and to the qualified scale-down model; the mechanism is characterized, not extrapolated, and movement beyond the characterized ranges would be assessed under the quality system (International Council for Harmonisation 2009).

## 6 Design space

The characterization defines a multivariate operating region in the well-controlled culture parameters over which the glycan, charge-variant and aggregate CQAs remain within their acceptance criteria across the characterized ranges. Because the governing factors act jointly rather than independently (§Results), that region is a multivariate envelope in the well-controlled CPPs, not a stack of one-dimensional ranges, and the response-surface models are its predictive basis.

The region’s principal planes are the factor pairs that carry the strongest interactions. The broad pair is culture pH  $\times$  culture duration, which governs the aggregate and afucosylation surfaces: both attributes rise toward the high-pH, long-duration corner, so that corner is the region’s worst case for size and afucosylation. The second plane is culture pH  $\times$  culture temperature, which governs high mannose — the very-high-criticality attribute — with pH raising it and temperature lowering it, so its binding corner is high-pH, low-temperature. Dissolved CO<sub>2</sub> forms a third, weaker plane through its dominant effect on the acidic charge variants. The operating region is drawn so that these worst-case corners remain within acceptance for every attribute across the ranges studied.

The normal operating ranges lie well inside this characterized region, which in turn lies inside the broader knowledge space bounded by the characterization ranges. Culture pH is controlled to a NOR of 6.75–6.95 inside a characterization range of 6.6–7.1; culture duration to 16–18 days inside 15–19 days; temperature to 34.5–35.5 °C inside 34–36 °C; and dissolved CO<sub>2</sub> to 70–130 mmHg inside 40–160 mmHg. Movement within the normal operating ranges is therefore not a change and requires no further assessment, whereas movement outside the characterized region would be assessed as a change under the quality system (CMC Biotech Working Group 2009; International Council for Harmonisation 2009).

The claim is bounded in three ways. It holds over the ranges studied and is not extrapolated beyond the characterization-range edges; it is a property of the four response-surface factors, with osmolality and the process-performance parameters held to their control ranges as robustness rather than region-defining factors; and it rests on the qualified scale-down model, so it is a commercial-equivalent claim to be confirmed by the process-performance-qualification campaign rather than a claim already demonstrated at manufacturing scale.

## 7 Proven acceptable ranges

Every well-controlled culture parameter’s proven acceptable range spans its entire characterization range — at set-point and under normal-operating-range co-variation alike — so no single parameter’s univariate range binds the operating region; the binding constraint is the multivariate design-space corner. Where the design space states the *joint* region in the four response-surface factors, a proven acceptable range (PAR) states, one parameter and one CQA at a time, how far that parameter may move before the CQA reaches its acceptance limit. The two views are consistent, and the reason they are is the finding of this section: because each parameter is individually robust across its whole studied span, what limits the region is the *combination* of parameters at a worst-case corner, not any one of them alone.

The acceptance criterion for each PAR is the study’s drug-substance specification for the CQA in question — the released-glycan, size-variant and charge-variant limits of Table 2.1 — applied as the ceiling, floor or two-sided window the attribute must meet. A single acceptance rule resolves both kinds of CQA: the drug-substance specification for an attribute the step forms, and, for a viral-clearance CQA, a back-calculated step floor — the step’s required log-reduction obtained from the cumulative requirement less the clearance credited to the other viral steps under the process’s modular-clearance framework. The production bioreactor forms no viral-clearance CQA, so that second case does not arise here; it is stated because the same rule governs the steps that do set those claims (PCR-006, PCR-008, PCR-009), and because the bioreactor’s own impurity CQAs are cleared downstream (PCR-005, PCR-007, PCR-008). Every PAR below is computed against the criterion that rule returns, so no acceptance limit is hand-carried into this section.

The proven acceptable ranges, computed for each of the 5 governed CQAs against each of the 4 response-surface parameters, are given in Table 7.1. Two ranges are reported per row. The *at-set-point* PAR is the interval over which the parameter may move with the other factors held at their set-points; the *NOR-propagated* PAR is the interval over which the fitted response-surface model keeps the 95 % predictive interval of the CQA within acceptance while the other factors vary across their normal operating ranges by Monte-Carlo — the more demanding of the two, and the one carried as the reproducible default because it charges each parameter’s latitude with the routine variability of the rest.

Table 7.1: Proven acceptable ranges at the production bioreactor — for each governed CQA and response-surface parameter, the characterization range and both proven acceptable ranges (at set-point and NOR-propagated).

| CQA           | Parameter           | Char. range | Unit | PAR (set-point) | PAR (NOR) |
|---------------|---------------------|-------------|------|-----------------|-----------|
| Afucosylation | Culture pH          | 6.6–7.1     | pH   | 6.6–7.1         | 6.6–7.1   |
| Afucosylation | Culture temperature | 34–36       | °C   | 34–36           | 34–36     |
| Afucosylation | Culture duration    | 15–19       | days | 15–19           | 15–19     |



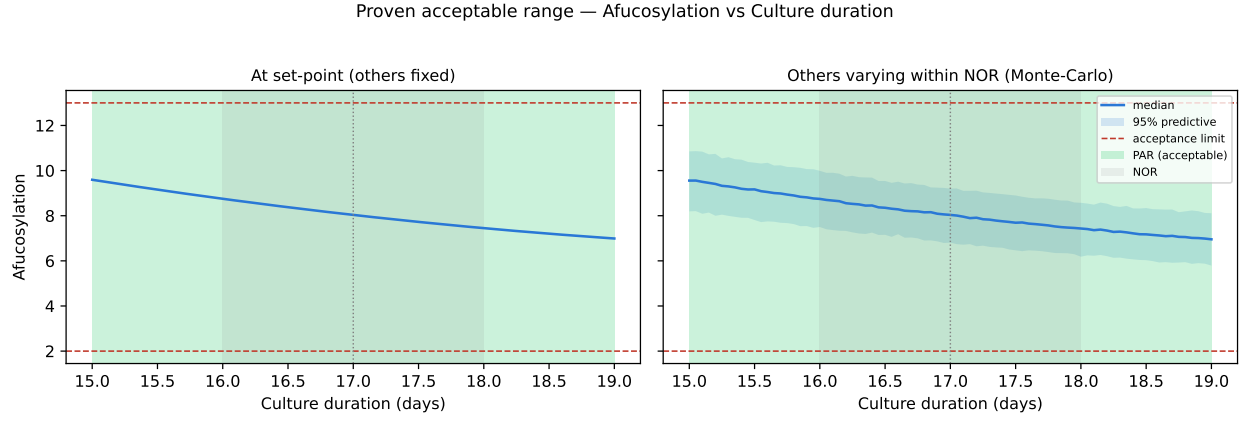
| CQA              | Parameter            | Char. range | Unit | PAR (set-point) | PAR (NOR) |
|------------------|----------------------|-------------|------|-----------------|-----------|
| Afucosylation    | Dissolved CO2 (pCO2) | 40–160      | mmHg | 40–160          | 40–160    |
| Galactosylation  | Culture pH           | 6.6–7.1     | pH   | 6.6–7.1         | 6.6–7.1   |
| Galactosylation  | Culture temperature  | 34–36       | °C   | 34–36           | 34–36     |
| Galactosylation  | Culture duration     | 15–19       | days | 15–19           | 15–19     |
| Galactosylation  | Dissolved CO2 (pCO2) | 40–160      | mmHg | 40–160          | 40–160    |
| High mannose     | Culture pH           | 6.6–7.1     | pH   | 6.6–7.1         | 6.6–7.1   |
| High mannose     | Culture temperature  | 34–36       | °C   | 34–36           | 34–36     |
| High mannose     | Culture duration     | 15–19       | days | 15–19           | 15–19     |
| High mannose     | Dissolved CO2 (pCO2) | 40–160      | mmHg | 40–160          | 40–160    |
| Acidic variants  | Culture pH           | 6.6–7.1     | pH   | 6.6–7.1         | 6.6–7.1   |
| Acidic variants  | Culture temperature  | 34–36       | °C   | 34–36           | 34–36     |
| Acidic variants  | Culture duration     | 15–19       | days | 15–19           | 15–19     |
| Acidic variants  | Dissolved CO2 (pCO2) | 40–160      | mmHg | 40–160          | 40–160    |
| Aggregates (HMW) | Culture pH           | 6.6–7.1     | pH   | 6.6–7.1         | 6.6–7.1   |
| Aggregates (HMW) | Culture temperature  | 34–36       | °C   | 34–36           | 34–36     |
| Aggregates (HMW) | Culture duration     | 15–19       | days | 15–19           | 15–19     |
| Aggregates (HMW) | Dissolved CO2 (pCO2) | 40–160      | mmHg | 40–160          | 40–160    |

Across all 20 CQA-by-parameter combinations both ranges coincide with the full characterization range (20 of 20 rows), which is the quantitative statement of univariate robustness: no single parameter, moved anywhere within its studied span, drives any of the 5 governed CQAs to its limit. That the NOR-propagated column reproduces the at-set-point column row for row is the stronger half of the claim — the latitude survives the remaining parameters co-varying across their NORs, not merely their being pinned at set-point. Read against the capability result, this is the same robustness seen from another direction: the tightest attribute clears its limit with wide margin (§Process capability and robustness), and here no parameter reaches that limit anywhere in its range.

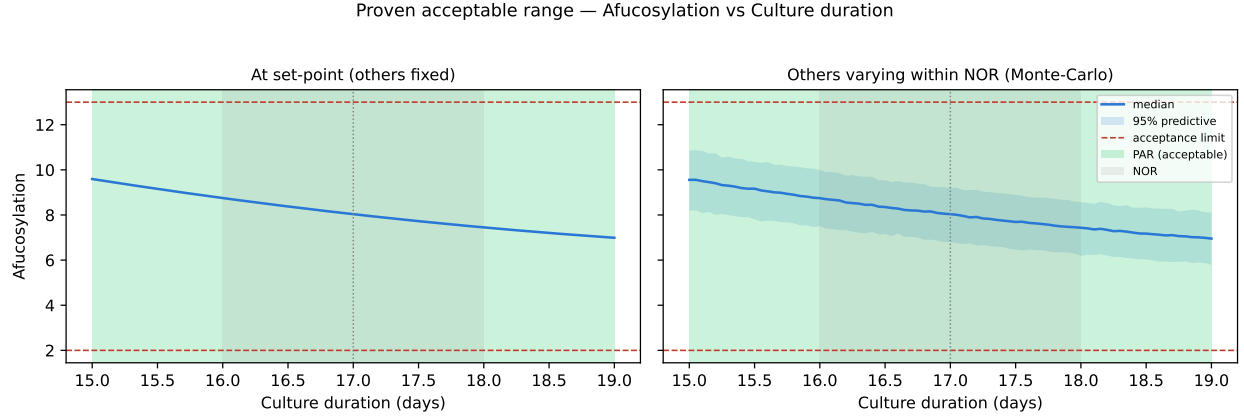
That every univariate range is fully acceptable is precisely why the operating region is bound multivariately rather than by any single parameter. High mannose — the very-high-criticality attribute with the tightest commercial capability — is governed by Culture pH, yet its PAR against Culture pH still spans the whole characterization range; the attribute approaches its limit only at the joint high-pH, low-temperature corner identified in the design space, not at any edge of a single parameter. Galactosylation behaves the same way: acceptable across the whole of every studied

parameter, it is steered toward its window only by the joint movement of its governing factors. The PAR analysis therefore neither widens nor replaces the design space; it demonstrates that the design-space corner, and not a one-dimensional range, is the active constraint on the operating region.

The per-CQA PAR plots make the table graphical: in each, the response over the governing parameter is drawn with its acceptance limit dashed, the acceptable parameter region shaded green, and the set-point and NOR marked, with the left panel holding the other factors at set-point and the right varying them within their NOR by Monte-Carlo. The efficacy-linked afucosylation and galactosylation glycans are both governed by Culture duration and rise gently with it, staying within their windows across the studied range (Figure 7.1, Figure 7.2).

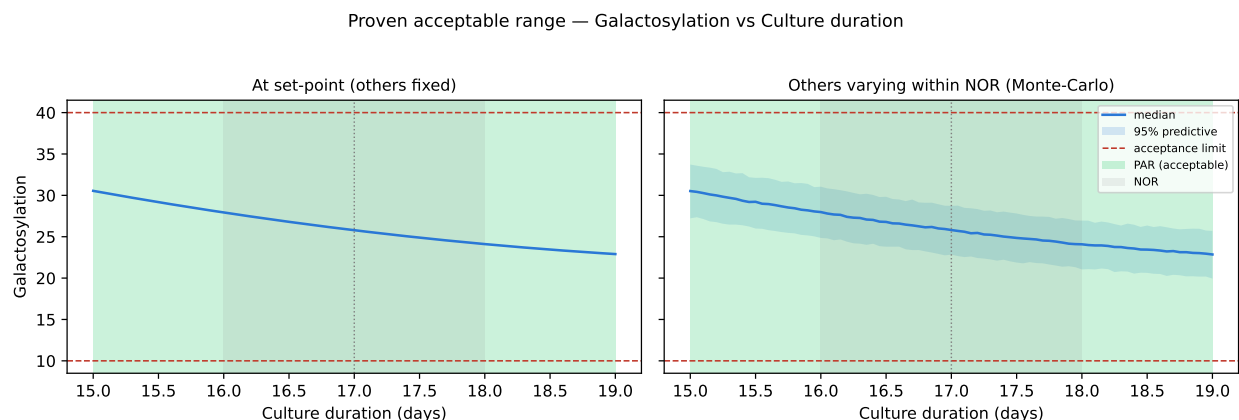


- (a) Proven acceptable range for afucosylation versus its governing parameter, at set-point (left) and with the other factors varying within their NOR (right); the acceptance limits are dashed, the acceptable region is shaded green, and the set-point and NOR are marked.

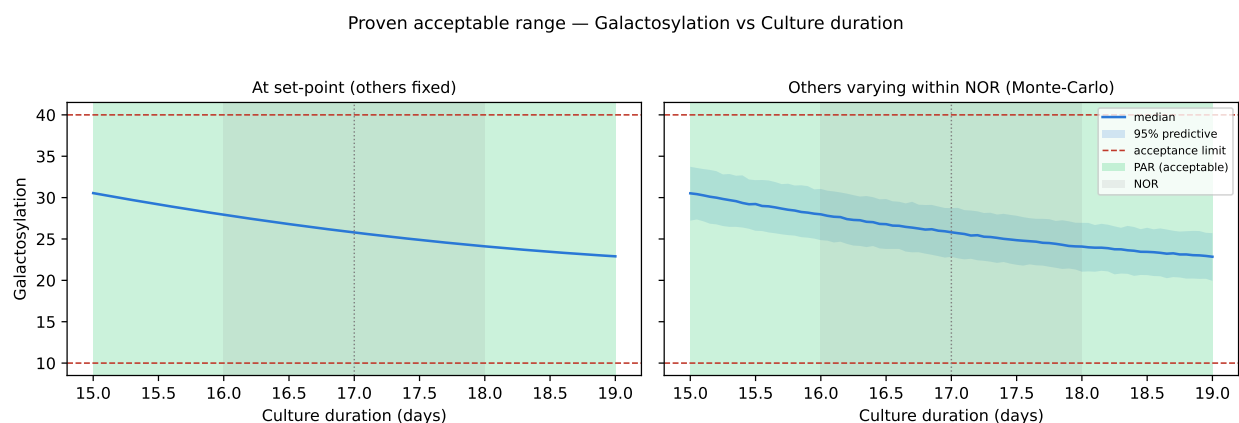


(b)

Figure 7.1



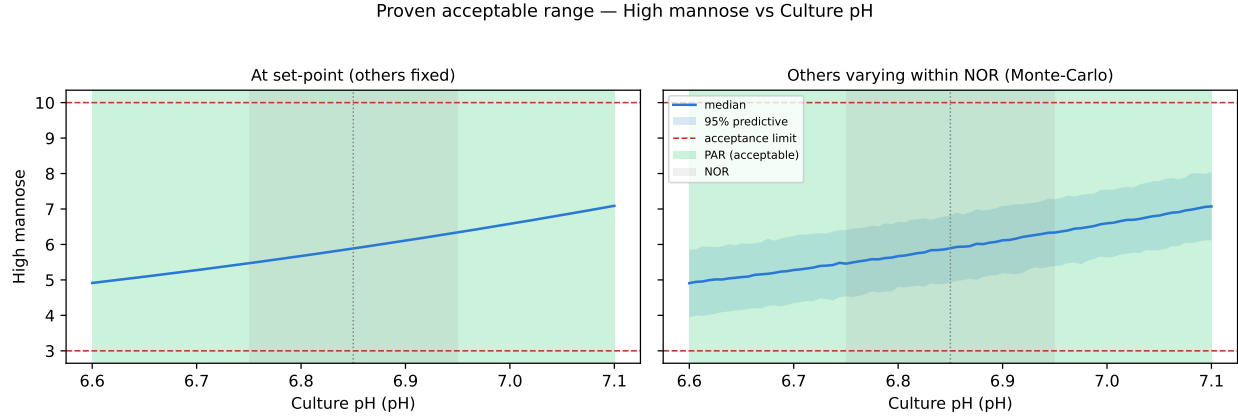
- (a) Proven acceptable range for galactosylation versus its governing parameter, at set-point (left) and NOR-propagated (right); acceptance window dashed, acceptable region shaded green, set-point and NOR marked.



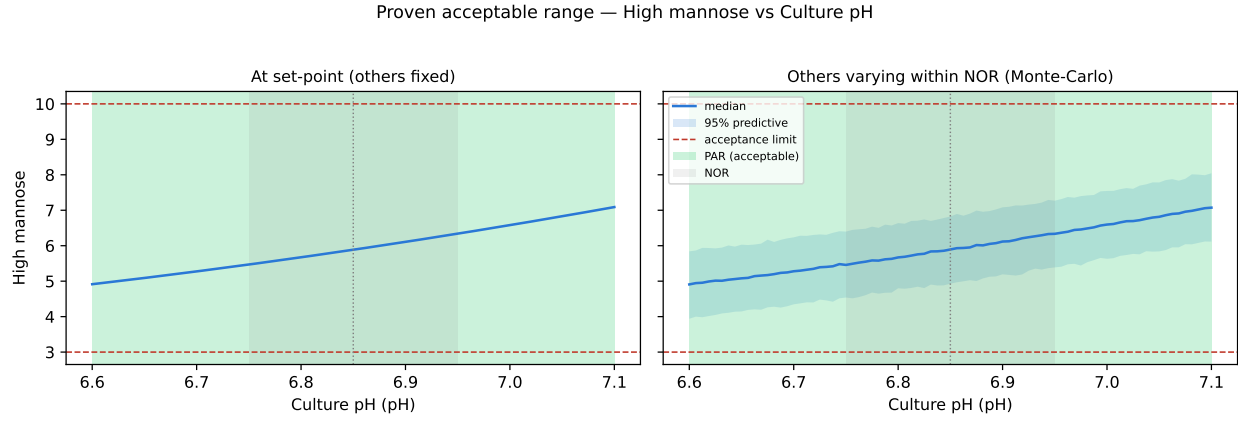
(b)

Figure 7.2

High mannose, the binding glycan, is governed by Culture pH, which raises it; across the whole characterization range of that parameter the response stays inside its two-sided window, and the NOR-propagated predictive band stays within acceptance (Figure 7.3).



- (a) Proven acceptable range for high mannose versus its governing parameter, at set-point (left) and NOR-propagated (right); the two-sided acceptance window is dashed, the acceptable region is shaded green, and the set-point and NOR are marked.

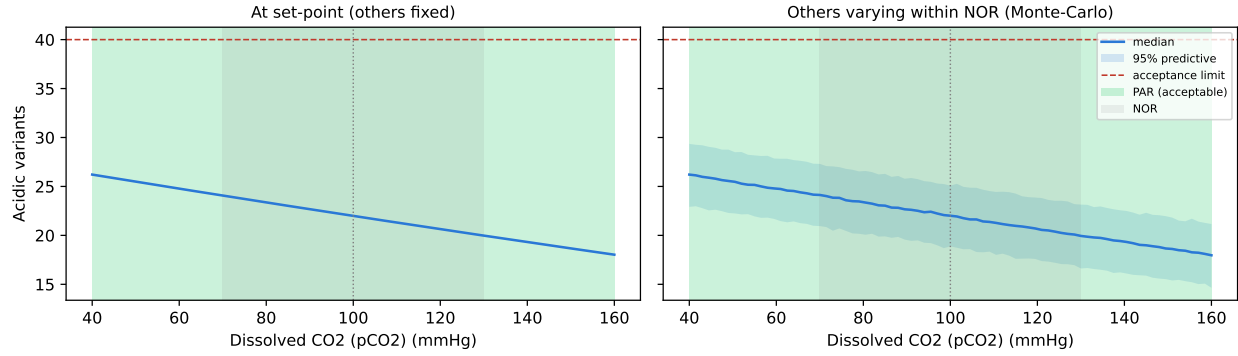


(b)

Figure 7.3

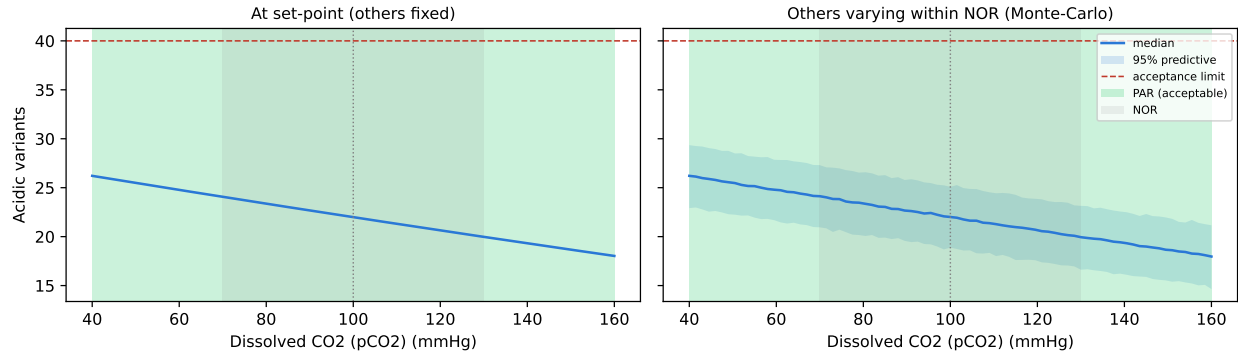
The acidic charge variants are governed by Dissolved CO<sub>2</sub> (pCO<sub>2</sub>) — their dominant driver — and stay below the upper acceptance limit across the full range of that parameter, consistent with the attribute's wide, clinically qualified window (Figure 7.4). The aggregate content, governed by Culture pH, likewise stays below its upper limit across the range (Figure 7.5).

Proven acceptable range — Acidic variants vs Dissolved CO<sub>2</sub> (pCO<sub>2</sub>)



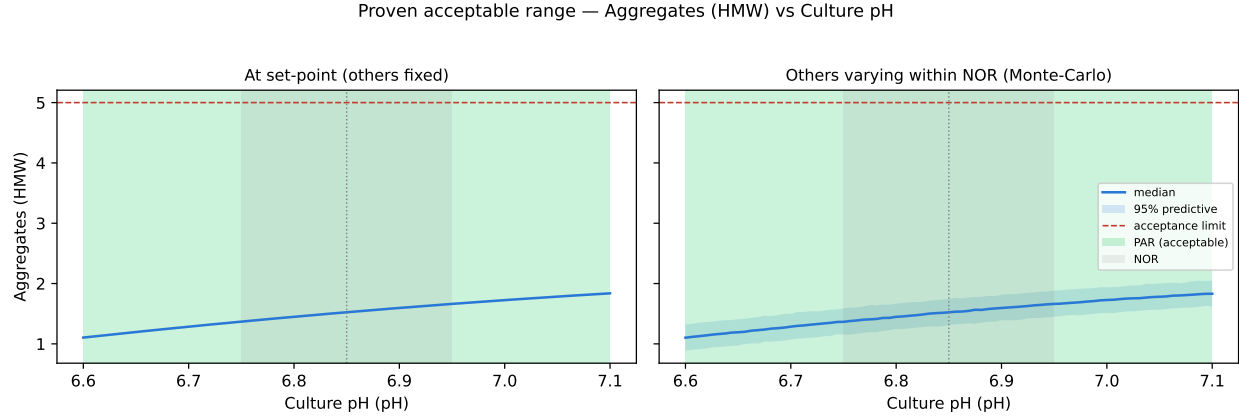
- (a) Proven acceptable range for the acidic charge variants versus their governing parameter, at set-point (left) and NOR-propagated (right); acceptance limit dashed, acceptable region shaded green, set-point and NOR marked.

Proven acceptable range — Acidic variants vs Dissolved CO<sub>2</sub> (pCO<sub>2</sub>)

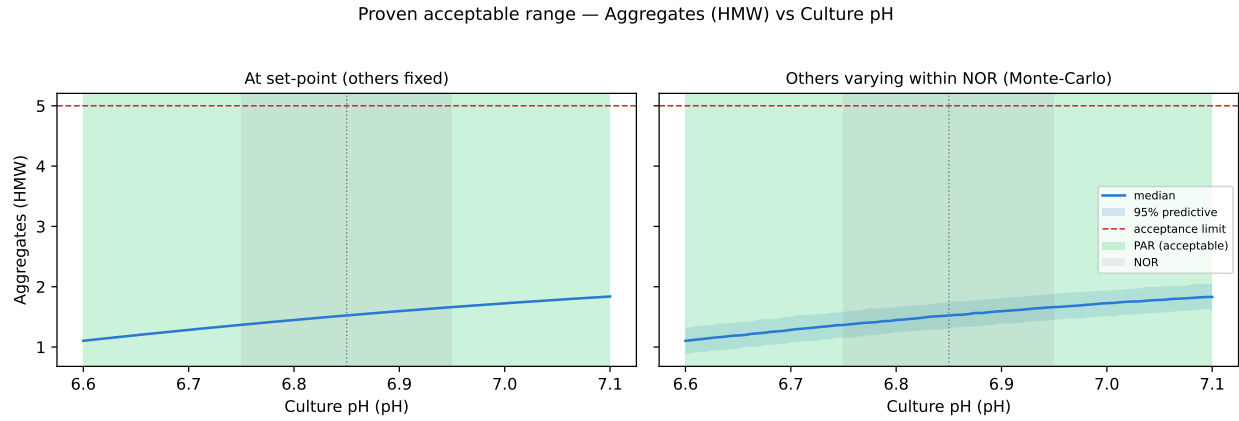


(b)

Figure 7.4



(a) Proven acceptable range for aggregates (HMW) versus its governing parameter, at set-point (left) and NOR-propagated (right); acceptance limit dashed, acceptable region shaded green, set-point and NOR marked.



(b)

Figure 7.5

These proven acceptable ranges are bounded in the same way as the design space they refine. They hold over the characterization ranges and are not extrapolated beyond them; a parameter's PAR is reported as its studied span because the response remained acceptable to that span's edges, not because acceptability was demonstrated further. They are properties of the fitted response-surface model on the qualified scale-down model, so the NOR-propagated ranges inherit that model's range of validity, and movement of any parameter beyond its characterization range would be assessed as a change under the quality system (International Council for Harmonisation 2009). Within those bounds the conclusion is unqualified: every studied parameter is proven acceptable across its whole characterization range, and the operating region is constrained by the multivariate design-space corner rather than by any univariate range.

## 8 Process capability and robustness

Every CQA the step forms is capable at commercial scale, with the tightest margin belonging to high mannose. A Monte-Carlo simulation of 2,000 batches, with the parameters varied across their normal operating ranges, propagated the culture variability into each response; the resulting one-sided capability indices are given in Table 8.1.

Table 8.1: Commercial-scale process capability for the CQAs formed at the production bioreactor, from the normal-operating-range Monte-Carlo simulation.

| CQA                                     | Crit. | Spec      | Acceptance | Mean $\pm$ SD     | Cpk  |
|-----------------------------------------|-------|-----------|------------|-------------------|------|
| Afucosylation                           | H     | two_sided | 2–13       | $7.98 \pm 0.58$   | 2.89 |
| Galactosylation (total %Gal)            | H     | two_sided | 10–40      | $26 \pm 1.6$      | 2.86 |
| High Mannose                            | VH    | two_sided | 3–10       | $5.99 \pm 0.51$   | 1.94 |
| Aggregates (HMW)                        | H     | upper     | 0–5        | $0.753 \pm 0.088$ | 16.1 |
| Acidic charge variants<br>(deamidation) | VL    | upper     | 18–40      | $22.3 \pm 1.4$    | 4.26 |
| Host Cell Protein (HCP)                 | M-H   | upper     | 0–100      | $15.2 \pm 4.6$    | 6.14 |
| Residual DNA                            | VL    | upper     | 0–0.001    | $0.0001 \pm 0$    | 10.8 |

The glycan attributes carry the binding capabilities (Table 8.1). High mannose is the tightest at Cpk 1.94 — about 5.8 standard deviations from its nearest acceptance limit — and its binding side is the lower limit (Cpl 1.94 against Cpu 2.60), consistent with its very-high criticality and with the larger measurement variance of a low-abundance glycan. Afucosylation and galactosylation are comfortably capable (Cpk 2.89 and 2.86), and the aggregate content is the most capable attribute of the set (Cpk 16.11), sitting far below its upper limit. The acidic charge variants are capable with wide margin (Cpk 4.26), as expected for a very-low-criticality attribute with a broad, clinically qualified range.

The host-cell-protein and residual-DNA capabilities (Cpk 6.14 and 10.81) are drug-substance capabilities that the bioreactor does not achieve alone. The bioreactor sets the incoming impurity burden; the margins reported here are realized only after the downstream clearance train — Protein A capture (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008) — has reduced that burden to release levels, and are consolidated on that basis in PCMR-001. The bioreactor’s own contribution is to deliver an incoming burden low and consistent enough for those steps to clear.

The capability claim is bounded to the normal operating ranges and to the qualified scale-down model: it is the predicted commercial-scale performance of a step operated within its NORs, to be confirmed by the process-performance-qualification batches, not a claim of demonstrated manufacturing-scale capability.

## 9 Parameter classification

Each parameter is classified from its demonstrated effect on the CQAs and the risk it would carry if it moved within its range; the classifications are summarized in Table 4.1 and justified individually below. 5 parameters are well-controlled critical process parameters, 3 are key process parameters, 1 is a general process parameter, and no parameter required the most stringent, critical-process-parameter control ( $n = 0$ ).

- **Culture pH — WC-CPP.** The most broadly active factor: it drives the aggregate and afucosylation surfaces (with duration) and the high-mannose surface (against temperature), recurring in the strongest interactions across responses. It is controlled to a narrow NOR of 6.75–6.95, within which the governed CQAs stay in limit, so it is a well-controlled — not an unqualified — critical parameter.
- **Culture temperature — WC-CPP.** Governs high mannose, which it lowers, and contributes to the aggregate response through its interaction with duration. It is held to 34.5–35.5 °C, a range over which high mannose remains within its very-high-criticality limits.
- **Dissolved CO (pCO) — WC-CPP.** The dominant driver of the acidic charge variants. Although that attribute is of very low criticality and wide capability, pCO is a credible, chemistry-linked driver of a CQA and is readily controlled, so it is classified WC-CPP and held to 70–130 mmHg.
- **Osmolality — WC-CPP.** Retained as a well-controlled critical parameter on the conservative platform basis that culture osmolality can influence glycosylation and cell physiology, even though it showed no dominant effect over the screened range. That null result is kept in the knowledge space as robustness evidence — it is why osmolality was not carried into the response-surface design — and the parameter is held to a narrow NOR of 375–425 mOsm.
- **Culture duration — WC-CPP.** Sets the time over which glycan processing and aggregation proceed and, with pH, governs the aggregate and afucosylation surfaces. It is controlled to 16–18 days, within which those attributes stay in limit.
- **Dissolved oxygen — KPP.** A process-performance parameter that supplies the culture's oxygen demand; over its range it showed no effect on the product-quality attributes, so it is controlled as a key parameter for culture performance rather than for a CQA, at 40–60 % saturation.
- **Initial viable cell concentration — KPP.** Assessed univariately; it acts through inoculation and early growth kinetics and did not move the product-quality attributes over its range, so it is a key process-performance parameter, controlled at 0.8–1.2 million cells/mL.
- **Nutrient feed-1 volume — KPP.** Supplies nutrients through the fed-batch profile; it is a process-performance parameter controlled at 10–14 % working volume. A single under-delivery on one centre-point batch (DEV-003-02) fell within this range and did not alter the fitted effects (§Deviations from the plan).
- **Basal medium concentration — GPP.** The lowest-risk parameter: over its wide range it showed no effect on product quality and carries no criticality, so it is a general process parameter, monitored rather than tightly controlled, at 1–1.4× the basal formulation.



No parameter met the threshold for critical-process-parameter control, because each active parameter is held to a range within which the CQAs it governs remain in limit and each inactive parameter is robust over its range; the well-controlled and key parameters together constitute the step's parameter control strategy.

## 10 Contribution to the control strategy

The production bioreactor contributes to the drug-substance control strategy by holding the culture parameters that form the product-quality attributes to the ranges characterized here, so that the incoming quality to the purification train is both in specification and consistent. The specific contributions are as follows. The five well-controlled critical parameters — culture pH, temperature, dissolved CO<sub>2</sub>, osmolality and duration — are controlled to their normal operating ranges (Table 4.1) with in-process monitoring of the culture environment, and it is these ranges that assure the glycan, charge-variant and aggregate CQAs. The key process parameters — dissolved oxygen, initial viable cell concentration and nutrient feed-1 volume — are controlled for culture performance, and the basal medium concentration is monitored as a general parameter. Product quality is verified on the end-of-culture material by the released-glycan, size-variant, charge-variant, HCP and DNA methods (Table 3.1). The feed-pump calibration control was reinforced following DEV-003-02 (§Deviations from the plan), a feed-input control arising from the campaign.

This step does not, by itself, deliver the process's impurity-clearance or viral-safety claims, and its control strategy is defined so as not to overreach that role. The host-cell-protein and residual-DNA burden it sets is cleared to release levels by Protein A capture (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008); the enveloped- and small-virus safety margins are established at the dedicated viral steps (PCR-006, PCR-008, PCR-009) under ICH Q5A(R2); and the glycan CQAs formed here are preserved, not altered, by the downstream steps. These contributions are consolidated with the present one in the Master Report (PCMR-001), which rolls up the 20 well-controlled critical parameters and the single critical process parameter ( $n = 1$ ) of the whole drug-substance process.

# 11 Discussion

The production bioreactor is well understood: the response-surface models reproduce the culture and its assays, they show no significant lack of fit against a demanding pure-error baseline, and they resolve the mechanism that platform prior knowledge predicts, so the operating region they define is a characterized region rather than an extrapolation. Confidence in the commercial relevance of that region rests on the scale-down model's qualification, which matched the small-scale culture to the manufacturing operation on both its inputs and its product-quality outputs; the tight centre-point reproducibility is the direct evidence that the model is representative at the set-point, and the factorial and axial runs extend that representativeness across the ranges.

Two deviations occurred during execution, and neither altered the study's conclusions. A dissolved-CO<sub>2</sub> probe drift on one screening run (DEV-003-01) read 6.5 mmHg high of the offline reference, and a nutrient-feed under-delivery on one centre-point batch (DEV-003-02) delivered 4.2 % less feed than target; the adverse findings are stated first because each touched a factor the study depends on — pCO<sub>2</sub>, the dominant charge-variant driver, and the centre point that anchors the pure-error estimate. In both cases the deviating value fell within the studied range and the fitted effects, the operating region and the classifications were shown to be unchanged (§Deviations from the plan), so both were retained.

The principal limitations are stated plainly, and each is managed forward. The claims are commercial-equivalent, resting on the scale-down model rather than on manufacturing-scale demonstration, and are to be confirmed by the process-performance-qualification campaign. The high-mannose capability, the tightest of the set, is limited in part by the measurement variance of a low-abundance glycan rather than by the culture, which is a reason to maintain the qualified assay as much as the culture control. And the design space is characterized only within the characterization ranges; operation beyond them would require assessment under the quality system. No parameter required critical-process-parameter control: every active parameter is a well-controlled critical parameter whose CQA risk is managed by range control, and the most stringent classification was therefore not needed at this step.

## 12 Conclusions

The production bioreactor is well characterized and robust within the ranges studied. A response-surface model over the well-controlled culture parameters defines a multivariate operating region in which the glycan, charge-variant and aggregate CQAs remain within their acceptance criteria; the proven acceptable and normal operating ranges are justified for each parameter; and the parameters are classified as 5 well-controlled critical, 3 key and 1 general process parameters, with no critical process parameter required. At commercial scale every CQA the step forms is capable, the tightest being high mannose at Cpk 1.94. The two recorded deviations were investigated, bounded and retained without altering any classification or the operating region. The step guarantees, on the qualified scale-down model and within the characterized ranges, the formation of in-specification product quality; it does not by itself deliver the impurity-clearance or viral-safety claims, which are established downstream and consolidated with this report in PCMR-001.

## 13 Deviations from the plan

2 deviations were recorded during execution of the study, and each was investigated to root cause, impact-assessed against the study conclusions and dispositioned; both were retained (Table 13.1). The two subsections below state the adverse finding first and then the evidence that bounds its impact.

Table 13.1: Deviation register for PCR-003 — the deviations recorded during execution, with their disposition.

| Devia-<br>tion | Summary                                                      | Detected during                            | Disposi-<br>tion |
|----------------|--------------------------------------------------------------|--------------------------------------------|------------------|
| DEV-003-01     | Dissolved-CO <sub>2</sub> probe drift during screening run 4 | in-process monitoring vs offline reference | retained         |
| DEV-003-02     | Nutrient feed-1 under-delivery on a centre-point batch       | post-execution mass reconciliation         | retained         |

### 13.1 DEV-003-01 — Dissolved-CO probe drift

During one screening run, in-process monitoring found the dissolved-CO probe (equipment EQ-BRX-205) reading 6.5 mmHg high of the offline reference; the root cause was probe drift between scheduled calibrations. This matters because dissolved CO is the single largest screening driver of the acidic charge variants (effect -8.2 % across the studied range), so the run's leverage on the pCO effect estimate was examined before the effects were accepted. The offset is small against the characterized pCO range of 40–160 mmHg, and the affected attribute is of very low criticality with wide commercial capability (Cpk 4.26); the pCO effect estimate, the response-surface model and the parameter classifications were unchanged, so the deviation was retained.

### 13.2 DEV-003-02 — Nutrient feed-1 under-delivery

Post-execution mass reconciliation found that one centre-point batch had received 4.2 % less nutrient feed-1 than target (feed lot LOT-FED-3120); the root cause was a feed-pump calibration offset. This matters because the centre-point replicates anchor the pure-error estimate against which every response-surface model's lack of fit is judged (§Results), so a non-representative centre point could distort that baseline. The batch's product-quality responses were therefore re-assessed by 3 verification measurements under AMV-3010, which placed it within the centre-point reproducibility band (95 % confidence half-width 6.21 on the affected glycan response). The delivered feed volume remained within the studied normal operating range of 10–14 % working volume, and the pure-error

estimate and the fitted effects were unchanged, so the deviation was retained; the feed-pump calibration control was reinforced for the commercial step (§Contribution to the control strategy).

# 14 Appendices

## 14.1 Appendix A — Screening design matrix and responses

The coded screening design matrix and its measured responses are given below, split into the factor settings (Table 14.1) and the responses (Table 14.2); the coded levels are  $-1$  and  $+1$  at the characterization-range edges and  $0$  at the centre.

Table 14.1: Screening design — coded factor settings.

| Run | Type      | A  | B  | C  | D  | E  |
|-----|-----------|----|----|----|----|----|
| 1   | factorial | -1 | -1 | -1 | -1 | 1  |
| 2   | factorial | -1 | -1 | -1 | 1  | -1 |
| 3   | factorial | -1 | -1 | 1  | -1 | -1 |
| 4   | factorial | -1 | -1 | 1  | 1  | 1  |
| 5   | factorial | -1 | 1  | -1 | -1 | -1 |
| 6   | factorial | -1 | 1  | -1 | 1  | 1  |
| 7   | factorial | -1 | 1  | 1  | -1 | 1  |
| 8   | factorial | -1 | 1  | 1  | 1  | -1 |
| 9   | factorial | 1  | -1 | -1 | -1 | -1 |
| 10  | factorial | 1  | -1 | -1 | 1  | 1  |
| 11  | factorial | 1  | -1 | 1  | -1 | 1  |
| 12  | factorial | 1  | -1 | 1  | 1  | -1 |
| 13  | factorial | 1  | 1  | -1 | -1 | 1  |
| 14  | factorial | 1  | 1  | -1 | 1  | -1 |
| 15  | factorial | 1  | 1  | 1  | -1 | -1 |
| 16  | factorial | 1  | 1  | 1  | 1  | 1  |
| 17  | center    | 0  | 0  | 0  | 0  | 0  |
| 18  | center    | 0  | 0  | 0  | 0  | 0  |
| 19  | center    | 0  | 0  | 0  | 0  | 0  |

Table 14.2: Screening design — measured responses.

| Run | Type      | Afucosylation | Galactosylation | High mannose | Acidic variants | Aggregates (HMW) |
|-----|-----------|---------------|-----------------|--------------|-----------------|------------------|
| 1   | factorial | 5.18          | 27.9            | 6.61         | 26.4            | 1.3              |
| 2   | factorial | 8.93          | 36.1            | 6.44         | 26.5            | 0.783            |
| 3   | factorial | 11.8          | 34.6            | 5.63         | 22.3            | 0.785            |
| 4   | factorial | 4.83          | 14.7            | 5.46         | 17.9            | 1.48             |
| 5   | factorial | 9.72          | 41.4            | 4.61         | 27              | 0.859            |

| Run | Type      | Afucosylation | Galactosylation | High mannose | Acidic variants | Aggregates (HMW) |
|-----|-----------|---------------|-----------------|--------------|-----------------|------------------|
| 6   | factorial | 4.68          | 27.3            | 4.68         | 27.6            | 1.26             |
| 7   | factorial | 6.22          | 14.5            | 4.65         | 23.1            | 1.48             |
| 8   | factorial | 9.83          | 31.2            | 4.45         | 19.3            | 0.948            |
| 9   | factorial | 9.61          | 28              | 7.77         | 24.1            | 1.51             |
| 10  | factorial | 7.51          | 17              | 7.89         | 20.4            | 2.26             |
| 11  | factorial | 7.13          | 19.5            | 7.34         | 12.6            | 2.53             |
| 12  | factorial | 5.29          | 17.6            | 7.01         | 11.3            | 1.61             |
| 13  | factorial | 6.74          | 28.3            | 6.43         | 29.6            | 2.18             |
| 14  | factorial | 5.4           | 26.7            | 5.94         | 24.4            | 1.62             |
| 15  | factorial | 3.82          | 25.2            | 6.1          | 18.4            | 1.8              |
| 16  | factorial | 3.08          | 29.1            | 6.15         | 15.8            | 2.05             |
| 17  | center    | 7.8           | 26.4            | 5.88         | 21.9            | 1.64             |
| 18  | center    | 7.51          | 27.4            | 5.84         | 20              | 1.63             |
| 19  | center    | 8.48          | 27              | 7.11         | 21.8            | 1.5              |

## 14.2 Appendix B — Response-surface design matrix and responses

The coded face-centred central-composite design and its responses are given below, split similarly into settings (Table 14.3) and responses (Table 14.4).

Table 14.3: Response-surface design — coded factor settings.

| Run | Type      | A  | B  | C  | E  |
|-----|-----------|----|----|----|----|
| 1   | factorial | -1 | -1 | -1 | -1 |
| 2   | factorial | -1 | -1 | 1  | -1 |
| 3   | factorial | -1 | -1 | -1 | 1  |
| 4   | factorial | -1 | -1 | 1  | 1  |
| 5   | factorial | -1 | 1  | -1 | -1 |
| 6   | factorial | -1 | 1  | 1  | -1 |
| 7   | factorial | -1 | 1  | -1 | 1  |
| 8   | factorial | -1 | 1  | 1  | 1  |
| 9   | factorial | 1  | -1 | -1 | -1 |
| 10  | factorial | 1  | -1 | 1  | -1 |
| 11  | factorial | 1  | -1 | -1 | 1  |
| 12  | factorial | 1  | -1 | 1  | 1  |
| 13  | factorial | 1  | 1  | -1 | -1 |
| 14  | factorial | 1  | 1  | 1  | -1 |
| 15  | factorial | 1  | 1  | -1 | 1  |
| 16  | factorial | 1  | 1  | 1  | 1  |
| 17  | axial     | -1 | 0  | 0  | 0  |
| 18  | axial     | 1  | 0  | 0  | 0  |
| 19  | axial     | 0  | -1 | 0  | 0  |
| 20  | axial     | 0  | 1  | 0  | 0  |



| Run | Type   | A | B | C  | E  |
|-----|--------|---|---|----|----|
| 21  | axial  | 0 | 0 | 0  | -1 |
| 22  | axial  | 0 | 0 | 0  | 1  |
| 23  | axial  | 0 | 0 | -1 | 0  |
| 24  | axial  | 0 | 0 | 1  | 0  |
| 25  | center | 0 | 0 | 0  | 0  |
| 26  | center | 0 | 0 | 0  | 0  |
| 27  | center | 0 | 0 | 0  | 0  |
| 28  | center | 0 | 0 | 0  | 0  |

Table 14.4: Response-surface design — measured responses.

| Run | Type      | Afucosylation | Galactosylation | High mannose | Acidic variants | Aggregates (HMW) |
|-----|-----------|---------------|-----------------|--------------|-----------------|------------------|
| 1   | factorial | 9.43          | 38.8            | 5.44         | 29.8            | 0.827            |
| 2   | factorial | 11.8          | 29              | 5.55         | 17.6            | 0.791            |
| 3   | factorial | 4.5           | 25.9            | 5.65         | 25.1            | 1.33             |
| 4   | factorial | 5.38          | 16.6            | 5.38         | 20              | 1.37             |
| 5   | factorial | 10.4          | 40.5            | 4.23         | 25.6            | 0.803            |
| 6   | factorial | 10.1          | 30.7            | 4.17         | 19.8            | 0.707            |
| 7   | factorial | 5.12          | 27.1            | 4.63         | 30.3            | 1.24             |
| 8   | factorial | 6.13          | 17.4            | 4.64         | 21.8            | 1.41             |
| 9   | factorial | 8.92          | 22              | 6.79         | 22.9            | 1.5              |
| 10  | factorial | 5.89          | 18.7            | 8.81         | 13.6            | 1.53             |
| 11  | factorial | 8.05          | 18.7            | 6.11         | 19.3            | 1.99             |
| 12  | factorial | 5.17          | 18.7            | 7.34         | 12.3            | 2.15             |
| 13  | factorial | 5.41          | 28.7            | 6.97         | 28.5            | 1.45             |
| 14  | factorial | 2.84          | 29.1            | 7.36         | 18.5            | 1.7              |
| 15  | factorial | 6.2           | 25.4            | 5.95         | 26.5            | 2.08             |
| 16  | factorial | 4.19          | 27.2            | 6.8          | 17.2            | 2.01             |
| 17  | axial     | 7.82          | 26.1            | 4.59         | 24.2            | 1.15             |
| 18  | axial     | 6.07          | 23.7            | 7.72         | 19.3            | 1.82             |
| 19  | axial     | 8.77          | 24.3            | 6.75         | 19.9            | 1.51             |
| 20  | axial     | 7.55          | 27.9            | 5.51         | 26.8            | 1.32             |
| 21  | axial     | 10.3          | 30.8            | 6.11         | 20.8            | 1.3              |
| 22  | axial     | 6.89          | 22.5            | 5.54         | 22.4            | 2.06             |
| 23  | axial     | 8.48          | 27.1            | 6.39         | 25.7            | 1.5              |
| 24  | axial     | 8.39          | 23.5            | 5.94         | 19.2            | 1.45             |
| 25  | center    | 7.88          | 23              | 5.75         | 20.5            | 1.52             |
| 26  | center    | 7.95          | 27.3            | 6.32         | 22              | 1.6              |
| 27  | center    | 7.35          | 25              | 5.6          | 22.4            | 1.55             |
| 28  | center    | 7.27          | 28.2            | 4.96         | 21.1            | 1.37             |

## 14.3 Appendix C — Full effect and coefficient tables

The complete screening effect estimates for all five responses, and the full quadratic response-surface coefficients for the four responses not shown in the Results section, are collected here.

Table 14.5: Screening effect estimates — Afucosylation (coded factors).

| Term | Effect | Coef.   | Std. err. | t      | p-value | Sig. |
|------|--------|---------|-----------|--------|---------|------|
| AE   | 2.46   | 1.23    | 0.266     | 4.62   | 0.0191  | *    |
| E    | -2.37  | -1.19   | 0.266     | -4.46  | 0.0209  | *    |
| AC   | -1.76  | -0.88   | 0.266     | -3.31  | 0.0455  | *    |
| A    | -1.57  | -0.785  | 0.266     | -2.95  | 0.06    |      |
| B    | -1.34  | -0.671  | 0.266     | -2.53  | 0.0858  |      |
| D    | -1.33  | -0.664  | 0.266     | -2.5   | 0.0878  |      |
| AB   | -1.28  | -0.64   | 0.266     | -2.41  | 0.0952  |      |
| C    | -0.73  | -0.365  | 0.266     | -1.37  | 0.264   |      |
| BD   | 0.452  | 0.226   | 0.266     | 0.85   | 0.458   |      |
| BE   | 0.362  | 0.181   | 0.266     | 0.681  | 0.545   |      |
| AD   | -0.175 | -0.0876 | 0.266     | -0.329 | 0.764   |      |
| BC   | -0.171 | -0.0856 | 0.266     | -0.322 | 0.769   |      |
| CD   | -0.147 | -0.0736 | 0.266     | -0.277 | 0.8     |      |
| DE   | 0.0358 | 0.0179  | 0.266     | 0.0674 | 0.951   |      |
| CE   | 0.0118 | 0.00591 | 0.266     | 0.0222 | 0.984   |      |

Table 14.6: Screening effect estimates — Galactosylation (coded factors).

| Term | Effect | Coef.   | Std. err. | t      | p-value  | Sig. |
|------|--------|---------|-----------|--------|----------|------|
| E    | -7.82  | -3.91   | 0.193     | -20.2  | 0.000264 | ***  |
| AE   | 6.9    | 3.45    | 0.193     | 17.9   | 0.000382 | ***  |
| C    | -5.79  | -2.9    | 0.193     | -15    | 0.000644 | ***  |
| A    | -4.56  | -2.28   | 0.193     | -11.8  | 0.00131  | **   |
| BD   | 3.7    | 1.85    | 0.193     | 9.57   | 0.00242  | **   |
| AC   | 3.6    | 1.8     | 0.193     | 9.32   | 0.00261  | **   |
| B    | 3.54   | 1.77    | 0.193     | 9.17   | 0.00274  | **   |
| AB   | 3.28   | 1.64    | 0.193     | 8.49   | 0.00343  | **   |
| D    | -2.45  | -1.23   | 0.193     | -6.35  | 0.0079   | **   |
| CD   | 2.18   | 1.09    | 0.193     | 5.64   | 0.011    | *    |
| DE   | 1.93   | 0.963   | 0.193     | 4.98   | 0.0155   | *    |
| BE   | 1.49   | 0.747   | 0.193     | 3.87   | 0.0306   | *    |
| AD   | -0.195 | -0.0974 | 0.193     | -0.504 | 0.649    |      |
| BC   | -0.131 | -0.0657 | 0.193     | -0.34  | 0.756    |      |
| CE   | 0.119  | 0.0596  | 0.193     | 0.309  | 0.778    |      |

Table 14.7: Screening effect estimates — High mannose (coded factors).

| Term | Effect    | Coef.     | Std. err. | t        | p-value | Sig. |
|------|-----------|-----------|-----------|----------|---------|------|
| A    | 1.51      | 0.756     | 0.155     | 4.89     | 0.0164  | *    |
| B    | -1.39     | -0.696    | 0.155     | -4.5     | 0.0205  | *    |
| C    | -0.446    | -0.223    | 0.155     | -1.44    | 0.245   |      |
| BC   | 0.369     | 0.185     | 0.155     | 1.19     | 0.318   |      |
| E    | 0.158     | 0.0791    | 0.155     | 0.511    | 0.645   |      |
| D    | -0.139    | -0.0693   | 0.155     | -0.448   | 0.685   |      |
| AC   | 0.0897    | 0.0449    | 0.155     | 0.29     | 0.791   |      |
| AE   | 0.0894    | 0.0447    | 0.155     | 0.289    | 0.792   |      |
| DE   | -0.0718   | -0.0359   | 0.155     | -0.232   | 0.831   |      |
| CE   | -0.0552   | -0.0276   | 0.155     | -0.179   | 0.87    |      |
| BE   | 0.046     | 0.023     | 0.155     | 0.149    | 0.891   |      |
| AB   | 0.0432    | 0.0216    | 0.155     | 0.14     | 0.898   |      |
| AD   | -0.0239   | -0.012    | 0.155     | -0.0774  | 0.943   |      |
| CD   | -0.0232   | -0.0116   | 0.155     | -0.075   | 0.945   |      |
| BD   | -0.000576 | -0.000288 | 0.155     | -0.00186 | 0.999   |      |

Table 14.8: Screening effect estimates — Acidic variants (coded factors).

| Term | Effect | Coef.  | Std. err. | t      | p-value  | Sig. |
|------|--------|--------|-----------|--------|----------|------|
| C    | -8.15  | -4.08  | 0.239     | -17    | 0.000441 | ***  |
| A    | -4.17  | -2.08  | 0.239     | -8.7   | 0.00319  | **   |
| B    | 2.97   | 1.48   | 0.239     | 6.19   | 0.00848  | **   |
| D    | -2.55  | -1.28  | 0.239     | -5.34  | 0.0129   | *    |
| AB   | 1.98   | 0.992  | 0.239     | 4.15   | 0.0255   | *    |
| AC   | -1.93  | -0.964 | 0.239     | -4.03  | 0.0275   | *    |
| BE   | 1.73   | 0.865  | 0.239     | 3.61   | 0.0364   | *    |
| AD   | -0.669 | -0.335 | 0.239     | -1.4   | 0.256    |      |
| CE   | -0.494 | -0.247 | 0.239     | -1.03  | 0.378    |      |
| CD   | -0.492 | -0.246 | 0.239     | -1.03  | 0.379    |      |
| BD   | -0.215 | -0.108 | 0.239     | -0.449 | 0.684    |      |
| BC   | 0.151  | 0.0757 | 0.239     | 0.316  | 0.773    |      |
| DE   | 0.0324 | 0.0162 | 0.239     | 0.0676 | 0.95     |      |
| E    | 0.0235 | 0.0117 | 0.239     | 0.049  | 0.964    |      |
| AE   | 0.0226 | 0.0113 | 0.239     | 0.0472 | 0.965    |      |

Table 14.9: Screening effect estimates — Aggregates (HMW) (coded factors).

| Term | Effect | Coef.   | Std. err. | t    | p-value  | Sig. |
|------|--------|---------|-----------|------|----------|------|
| A    | 0.834  | 0.417   | 0.0207    | 20.1 | 0.000268 | ***  |
| E    | 0.578  | 0.289   | 0.0207    | 13.9 | 0.0008   | ***  |
| BE   | -0.141 | -0.0705 | 0.0207    | -3.4 | 0.0424   | *    |

| Term | Effect   | Coef.    | Std. err. | t      | p-value | Sig. |
|------|----------|----------|-----------|--------|---------|------|
| C    | 0.116    | 0.058    | 0.0207    | 2.8    | 0.0678  |      |
| CD   | -0.0742  | -0.0371  | 0.0207    | -1.79  | 0.171   |      |
| AD   | -0.0666  | -0.0333  | 0.0207    | -1.61  | 0.206   |      |
| AB   | -0.0593  | -0.0297  | 0.0207    | -1.43  | 0.248   |      |
| DE   | -0.0554  | -0.0277  | 0.0207    | -1.34  | 0.274   |      |
| D    | -0.0548  | -0.0274  | 0.0207    | -1.32  | 0.278   |      |
| BD   | -0.0547  | -0.0274  | 0.0207    | -1.32  | 0.278   |      |
| AE   | 0.0416   | 0.0208   | 0.0207    | 1      | 0.39    |      |
| BC   | -0.0235  | -0.0118  | 0.0207    | -0.567 | 0.61    |      |
| CE   | 0.0222   | 0.0111   | 0.0207    | 0.537  | 0.629   |      |
| B    | -0.00671 | -0.00336 | 0.0207    | -0.162 | 0.882   |      |
| AC   | -0.00663 | -0.00332 | 0.0207    | -0.16  | 0.883   |      |

Table 14.10: Response-surface coefficients — Galactosylation (coded factors).

| Term           | Coef.  | Std. err. | t      | p-value  | Sig. |
|----------------|--------|-----------|--------|----------|------|
| Intercept      | 25.8   | 0.47      | 54.8   | 9.11e-17 | ***  |
| A              | -2.21  | 0.321     | -6.89  | 1.09e-05 | ***  |
| B              | 2.3    | 0.321     | 7.18   | 7.13e-06 | ***  |
| E              | -3.82  | 0.321     | -11.9  | 2.27e-08 | ***  |
| C              | -2.43  | 0.321     | -7.56  | 4.11e-06 | ***  |
| AB             | 1.67   | 0.34      | 4.92   | 0.000279 | ***  |
| AE             | 2.72   | 0.34      | 7.99   | 2.28e-06 | ***  |
| AC             | 2.35   | 0.34      | 6.9    | 1.09e-05 | ***  |
| BE             | -0.217 | 0.34      | -0.639 | 0.534    |      |
| BC             | 0.315  | 0.34      | 0.928  | 0.37     |      |
| EC             | 0.332  | 0.34      | 0.975  | 0.347    |      |
| A <sup>2</sup> | -0.837 | 0.847     | -0.988 | 0.341    |      |
| B <sup>2</sup> | 0.378  | 0.847     | 0.446  | 0.663    |      |
| E <sup>2</sup> | 0.937  | 0.847     | 1.11   | 0.289    |      |
| C <sup>2</sup> | -0.373 | 0.847     | -0.441 | 0.667    |      |

Table 14.11: Response-surface coefficients — High mannose (coded factors).

| Term      | Coef.  | Std. err. | t     | p-value  | Sig. |
|-----------|--------|-----------|-------|----------|------|
| Intercept | 5.89   | 0.163     | 36    | 2.06e-14 | ***  |
| A         | 1.09   | 0.111     | 9.77  | 2.36e-07 | ***  |
| B         | -0.421 | 0.111     | -3.78 | 0.00227  | **   |
| E         | -0.188 | 0.111     | -1.69 | 0.115    |      |
| C         | 0.212  | 0.111     | 1.91  | 0.0788   |      |
| AB        | 0.148  | 0.118     | 1.26  | 0.231    |      |
| AE        | -0.29  | 0.118     | -2.46 | 0.0289   | *    |
| AC        | 0.293  | 0.118     | 2.48  | 0.0276   | *    |

| Term           | Coef.   | Std. err. | t      | p-value | Sig. |
|----------------|---------|-----------|--------|---------|------|
| BE             | 0.0874  | 0.118     | 0.74   | 0.472   |      |
| BC             | -0.118  | 0.118     | -0.996 | 0.337   |      |
| EC             | -0.0402 | 0.118     | -0.341 | 0.739   |      |
| A <sup>2</sup> | 0.116   | 0.294     | 0.393  | 0.701   |      |
| B <sup>2</sup> | 0.094   | 0.294     | 0.32   | 0.754   |      |
| E <sup>2</sup> | -0.215  | 0.294     | -0.731 | 0.478   |      |
| C <sup>2</sup> | 0.126   | 0.294     | 0.43   | 0.674   |      |

Table 14.12: Response-surface coefficients — Acidic variants (coded factors).

| Term           | Coef.   | Std. err. | t       | p-value  | Sig. |
|----------------|---------|-----------|---------|----------|------|
| Intercept      | 22      | 0.542     | 40.6    | 4.43e-15 | ***  |
| A              | -2.01   | 0.369     | -5.44   | 0.000113 | ***  |
| B              | 1.9     | 0.369     | 5.14    | 0.000191 | ***  |
| E              | -0.124  | 0.369     | -0.335  | 0.743    |      |
| C              | -4.09   | 0.369     | -11.1   | 5.4e-08  | ***  |
| AB             | 1.11    | 0.392     | 2.84    | 0.0139   | *    |
| AE             | -0.793  | 0.392     | -2.03   | 0.0638   |      |
| AC             | -0.248  | 0.392     | -0.633  | 0.538    |      |
| BE             | 0.655   | 0.392     | 1.67    | 0.118    |      |
| BC             | -0.0111 | 0.392     | -0.0283 | 0.978    |      |
| EC             | 0.465   | 0.392     | 1.19    | 0.256    |      |
| A <sup>2</sup> | -0.586  | 0.975     | -0.601  | 0.558    |      |
| B <sup>2</sup> | 1.03    | 0.975     | 1.06    | 0.311    |      |
| E <sup>2</sup> | -0.718  | 0.975     | -0.736  | 0.475    |      |
| C <sup>2</sup> | 0.13    | 0.975     | 0.133   | 0.896    |      |

Table 14.13: Response-surface coefficients — Aggregates (HMW) (coded factors).

| Term           | Coef.    | Std. err. | t      | p-value  | Sig. |
|----------------|----------|-----------|--------|----------|------|
| Intercept      | 1.52     | 0.0341    | 44.8   | 1.25e-15 | ***  |
| A              | 0.367    | 0.0232    | 15.8   | 7.26e-10 | ***  |
| B              | -0.0157  | 0.0232    | -0.675 | 0.512    |      |
| E              | 0.281    | 0.0232    | 12.1   | 1.91e-08 | ***  |
| C              | 0.0225   | 0.0232    | 0.971  | 0.349    |      |
| AB             | 0.0135   | 0.0246    | 0.547  | 0.593    |      |
| AE             | -0.0106  | 0.0246    | -0.43  | 0.674    |      |
| AC             | 0.0179   | 0.0246    | 0.727  | 0.48     |      |
| BE             | -0.00713 | 0.0246    | -0.29  | 0.777    |      |
| BC             | 0.00414  | 0.0246    | 0.168  | 0.869    |      |
| EC             | 0.0105   | 0.0246    | 0.426  | 0.677    |      |
| A <sup>2</sup> | -0.0533  | 0.0613    | -0.87  | 0.4      |      |
| B <sup>2</sup> | -0.119   | 0.0613    | -1.93  | 0.0752   |      |

| Term           | Coef.   | Std. err. | t     | p-value | Sig. |
|----------------|---------|-----------|-------|---------|------|
| E <sup>2</sup> | 0.143   | 0.0613    | 2.33  | 0.0366  | *    |
| C <sup>2</sup> | -0.0626 | 0.0613    | -1.02 | 0.326   |      |

## 14.4 Appendix D — Analytical methods summary

The validated analytical methods used to measure the characterization responses are summarized in Table 14.14; method performance is detailed in the respective method-validation reports.

Table 14.14: Analytical methods summary — technique, analyte and reportable unit for each validated method.

| Method validation | Method / technique             | Analyte(s)                                   | Reportable unit |
|-------------------|--------------------------------|----------------------------------------------|-----------------|
| AMV-3010          | N-Glycan Map (2-AB HILIC-UPLC) | Afucosylation, galactosylation, high mannose | % of glycans    |
| AMV-3011          | Size-Variants (SEC-HPLC)       | Aggregates (HMW)                             | % HMW (SEC)     |
| AMV-3013          | Charge Variants (icIEF)        | Acidic charge variants (deamidation)         | % (CEX/icIEF)   |
| AMV-3012          | Host-Cell Protein ELISA        | Host cell protein (HCP)                      | ng/mg           |
| AMV-3014          | Residual DNA (qPCR)            | Residual DNA                                 | ng/dose         |

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